



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

**SCHOOL OF COMPUTING**  
Faculty of Engineering

## **Semester II 2021/2022**

**Subject : Bioinformatics I (SECB2103)**  
**Section : 01 – Dr Haslina Hashim**  
**Topic : Lab 06 – Multiple Sequence Analysis**

**Name : Koo Xuan (A23CS0300)**

### **1) Practice using three NCBI resources to obtain groups of sequences in the FASTA format that you can use for multiple sequence alignment.**

#### **FIRST APPROACH:**

- i) Select a keyword such as cytochrome (other suggestions are ferritin, S100 or trypsin).  
In a first approach, enter this search from the home page of NCBI, and follow the link to HomoloGene. By default, the entries are displayed in the summary
- ii) Using the pull-down menu change the display to Multiple Alignment. This allows you to scroll through a series of multiple sequence alignments. Select one for further study. It is helpful to choose one in which there are some gaps.
- iii) Once you identify a group of proteins of interest, click to view that HomoloGene group, and change the display to FASTA.
- iv) Copy these sequences and/or save them to a text document.

#### **SECOND APPROACH:**

- i) Repeat this exercise beginning at the home page of NCBI, but select the link to CDD (the conserved domain database). Here, there are pfam, cdd, smart, and/or COG identifiers.
- ii) Select an entry with a CDD identifier (such as cd00904 for ferritin). Here, a multiple sequence alignment is shown.
- iii) Change the format to obtain the desired number of proteins in this family (e.g. up to 5, 10, or 20) in the FASTA format; you may select the most diverse members of this group.

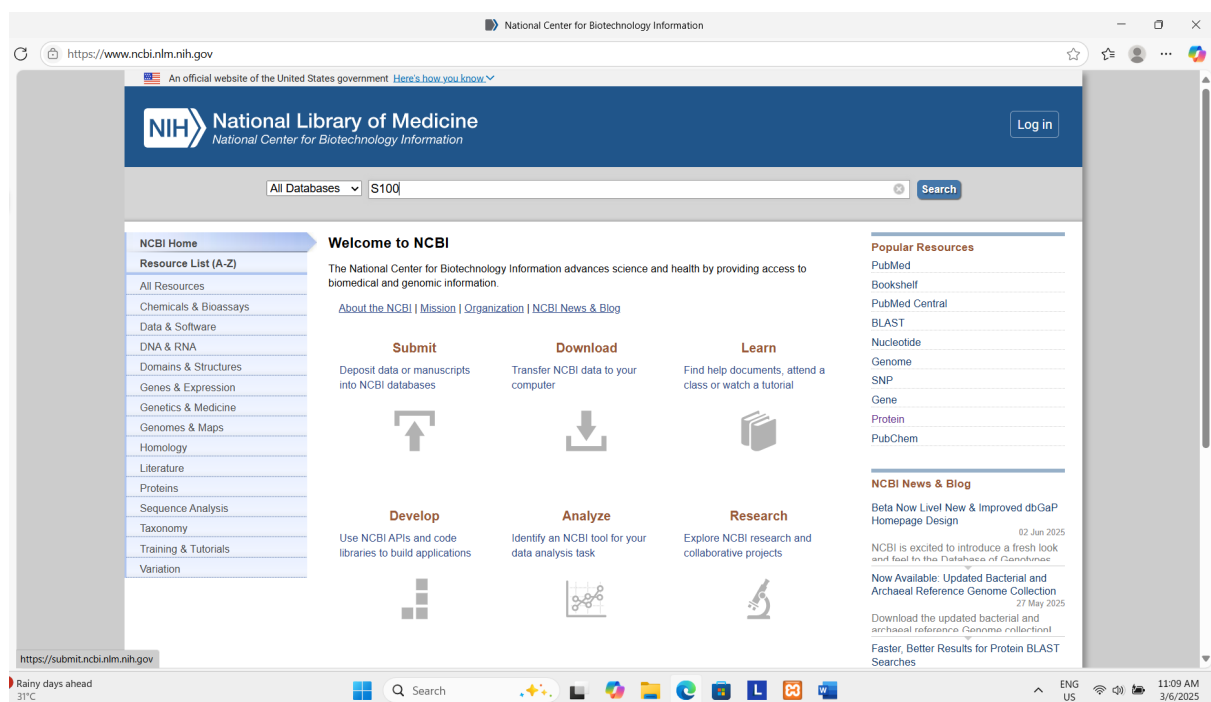
### THIRD APPROACH:

- i) Perform a BLASTP search using a query such as ferritin light chain (NP\_000137) and inspect the pairwise alignments to the query.
- ii) Select a group of ten proteins by clicking on the box next to each, and click “Get selected sequences.”
- iii) These ten protein appear on an NCBI Protein page; change the display option to FASTA and use the pull-down menu option “send to text.” The sequences are now available in the FASTA format for further study.

### (1) SOLUTIONS/COMMENTS:

#### FIRST APPROACH:

To begin with first approach, we begin at the home page of NCBI with a search for S100:



So here the results for S100, and include S100A1 as the second entry.

Gene S100 Search

Gene sources: Genomic, Categories, Sequence content, Status, **Current**

Search results: 1 to 20 of 16783

| Name/Gene ID                                           | Description                                                                               | Location                                                    | Aliases                                             | MIM    |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|--------|
| <input type="checkbox"/> S100B<br>ID: 6285             | S100 calcium binding protein B [ <i>Homo sapiens</i> (human)]                             | Chromosome 21, NC_000021.9 (46598604..46605082, complement) | NEF, S100, S100-B, S100beta                         | 176990 |
| <input type="checkbox"/> S100A1<br>ID: 6271            | S100 calcium binding protein A1 [ <i>Homo sapiens</i> (human)]                            | Chromosome 1, NC_000001.11 (153628434..153632039)           | S100, S100-alpha, S100A                             | 176940 |
| <input type="checkbox"/> S100a1<br>ID: 20193           | S100 calcium binding protein A1 [ <i>Mus musculus</i> (house mouse)]                      | Chromosome 3, NC_000069.7 (90418341..90421637, complement)  | S100, S100a                                         |        |
| <input type="checkbox"/> LOC100136538<br>ID: 100136538 | protein S100-A1 [ <i>Salmo salar</i> (Atlantic salmon)]                                   | Chromosome ssa02, NC_059443.1 (18253086..18280834)          | S100-A1, S10A1, s100                                |        |
| <input type="checkbox"/> s100a1.L<br>ID: 108699648     | S100 calcium binding protein A1 L homeolog [ <i>Xenopus laevis</i> (African clawed frog)] | Chromosome 8L, NC_054385.1 (133231864..133249758)           | XELAEV_18040853mg, s100, s100-a1, s100-alpha, s100a |        |
| <input type="checkbox"/> s100a1<br>ID: 101734724       | S100 calcium binding protein A1 [ <i>Xenopus tropicalis</i> (tropical clawed frog)]       | Chromosome 8, NC_030684.2 (144085357..144107338)            | s100, s100-a1, s100-alpha, s100a                    |        |

Filters: Manage Filters

Results by taxon: Top Organisms [Tree]  
Homo sapiens (152)  
Salmonella phage S100 (67)  
Elephantulus edwardii (60)  
Nycticebus coucang (50)  
Mus musculus (50)  
All other taxa (16388)

Find related data: Database: [Select]  
Find items

Search details: S100[All Fields] AND alive[prop]  
Search See more...

Recent activity: S100 AND (alive[prop]) (16783)  
Turn Off Clear

Scroll down and find for orthologs, and click all

Full Report S100A1 S100 calcium binding protein A1 [*Homo sapiens* (human)]

Gene ID: 6271, updated on 24-May-2025

Download Datasets

Summary

Official Symbol: S100A1 provided by HGNC  
Official Full Name: S100 calcium binding protein A1 provided by HGNC  
Primary source: HGNC:HGNC:10486  
See related: Ensembl:ENS00000160678 MIM:176940 AllianceGenome:HGNC:10486  
Gene type: protein coding  
RefSeq status: REVIEWED  
Organism: [Homo sapiens](#)  
Lineage: Eukaryota, Metazoa, Chordata, Craniata, Vertebrata, Euteleostomi, Mammalia, Eutheria, Euarchontoglires, Primates, Haplorhini, Catarrhini, Hominidae, Homo  
Also known as: S100, S100A, S100-alpha  
Summary: The protein encoded by this gene is a member of the S100 family of proteins containing 2 EF-hand calcium-binding motifs. S100 proteins are localized in the cytoplasm and/or nucleus of a wide range of cells, and involved in the regulation of a number of cellular processes such as cell cycle progression and differentiation. S100 genes include at least 13 members which are located as a cluster on chromosome 1q21. This protein may function in stimulation of Ca2+-induced Ca2+ release, inhibition of microtubule assembly, and inhibition of protein kinase C-mediated phosphorylation. Reduced expression of this protein has been implicated in cardiomyopathies. [provided by RefSeq, Jul 2008]  
Expression: Biased expression in heart (RPKM 773.3), salivary gland (RPKM 114.1) and 3 other tissues [See more](#)  
Orthologs: [mouse](#) [all](#)  
**NEW** Try the new [Gene table](#)  
Try the new [Transcript table](#)

Genomic context

Location: 1q21.3 See S100A1 in [Genome Data Viewer](#)  
Exon count: 3

| Annotation release                                                                                              | Status  | Assembly                      | Chr | Location                            |
|-----------------------------------------------------------------------------------------------------------------|---------|-------------------------------|-----|-------------------------------------|
| RS_2024_08                                                                                                      | current | GRCh38.p14 (GCF_000001405.40) | 1   | NC_000001.11 (153628434..153632039) |
| <a href="https://www.ncbi.nlm.nih.gov/gene/6271/ortholog/">https://www.ncbi.nlm.nih.gov/gene/6271/ortholog/</a> |         |                               |     |                                     |

Table of contents: Summary, Genomic context, Expression, Bibliography, Phenotypes, Variation, HIV-1 interactions, Pathways from PubChem, Interactions, General gene information, General protein information, NCBI Reference Sequences (RefSeq), Related sequences, Additional links

Genome Browsers: Genome Data Viewer, Variation Viewer (GRCh37.p13), Variation Viewer (GRCh38), Ensembl, UCSC

And it will direct to a comparative genomics view. It displays a list of orthologous genes in different species and select any three of them.

https://www.ncbi.nlm.nih.gov/gene/6271/ortholog/?scope=7742

Search NCBI

### S100A1 - S100 calcium binding protein A1

The protein encoded by this gene is a member of the S100 family of proteins containing 2 EF-hand calcium-binding motifs. S100 proteins are localized in the cytoplasm and/or nucleus of a wide range of cells, and involved in the regulation of a number of cellular processes such as cell cycle progression and differentiation. S100 genes include at least 13 members which are located as a cluster on chromosome 1q21. This protein may function in stimulation of Ca<sup>2+</sup>-induced Ca<sup>2+</sup> release, inhibition of microtubule assembly, and inhibition of protein kinase C-mediated phosphorylation. Reduced expression of this protein has been implicated in cardiomyopathies. [provided by RefSeq, Jul 2008]

Genes similar to S100A1

#### NCBI Orthologs

How was this calculated?

0 items

406 genes for: vertebrates (Vertebrata)

SEARCH THE TAXONOMY TREE

Enter taxonomic name

- vertebrates
  - birds
  - turtles
  - alligators and others
  - lizards & snakes
  - mammals
  - amphibians
  - lampreys

3 selected

| Species                                                                    | Gene                                      | Architecture | aa  | Previous | Next |
|----------------------------------------------------------------------------|-------------------------------------------|--------------|-----|----------|------|
| <input checked="" type="checkbox"/> <i>Homo sapiens</i><br>human           | S100A1<br>S100 calcium binding protein A1 |              | 94  |          |      |
| <input checked="" type="checkbox"/> <i>Mus musculus</i><br>house mouse     | S100a1<br>S100 calcium binding protein A1 |              | 94  |          |      |
| <input checked="" type="checkbox"/> <i>Rattus norvegicus</i><br>Norway rat | S100a1<br>S100 calcium binding protein A1 |              | 94  |          |      |
| <input type="checkbox"/> <i>Bos taurus</i><br>domestic cattle              | S100A1<br>S100 calcium binding protein A1 |              | 187 |          |      |
| <input type="checkbox"/> <i>Macaca mulatta</i>                             | S100A1                                    |              | 94  |          |      |

Feedback

Download data for the all sequence per gene in FASTA format.

https://www.ncbi.nlm.nih.gov/gene/6271/ortholog/?scope=7742

### S100A1 - S100 calcium binding protein A1

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Genes similar to S100A1

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SEARCH THE TAXONOMY TREE

Enter taxonomic name

- vertebrates
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  - turtles
  - alligators and others
  - lizards & snakes
  - mammals
  - amphibians
  - lampreys

3 selected

| Species                                                                    | Gene                                      | Architecture | aa  | Previous | Next |
|----------------------------------------------------------------------------|-------------------------------------------|--------------|-----|----------|------|
| <input checked="" type="checkbox"/> <i>Homo sapiens</i><br>human           | S100A1<br>S100 calcium binding protein A1 |              | 94  |          |      |
| <input checked="" type="checkbox"/> <i>Mus musculus</i><br>house mouse     | S100a1<br>S100 calcium binding protein A1 |              | 94  |          |      |
| <input checked="" type="checkbox"/> <i>Rattus norvegicus</i><br>Norway rat | S100a1<br>S100 calcium binding protein A1 |              | 94  |          |      |
| <input type="checkbox"/> <i>Bos taurus</i><br>domestic cattle              | S100A1<br>S100 calcium binding protein A1 |              | 187 |          |      |

Download data

File types

RefSeq transcripts (FASTA)

☐ one sequence per gene

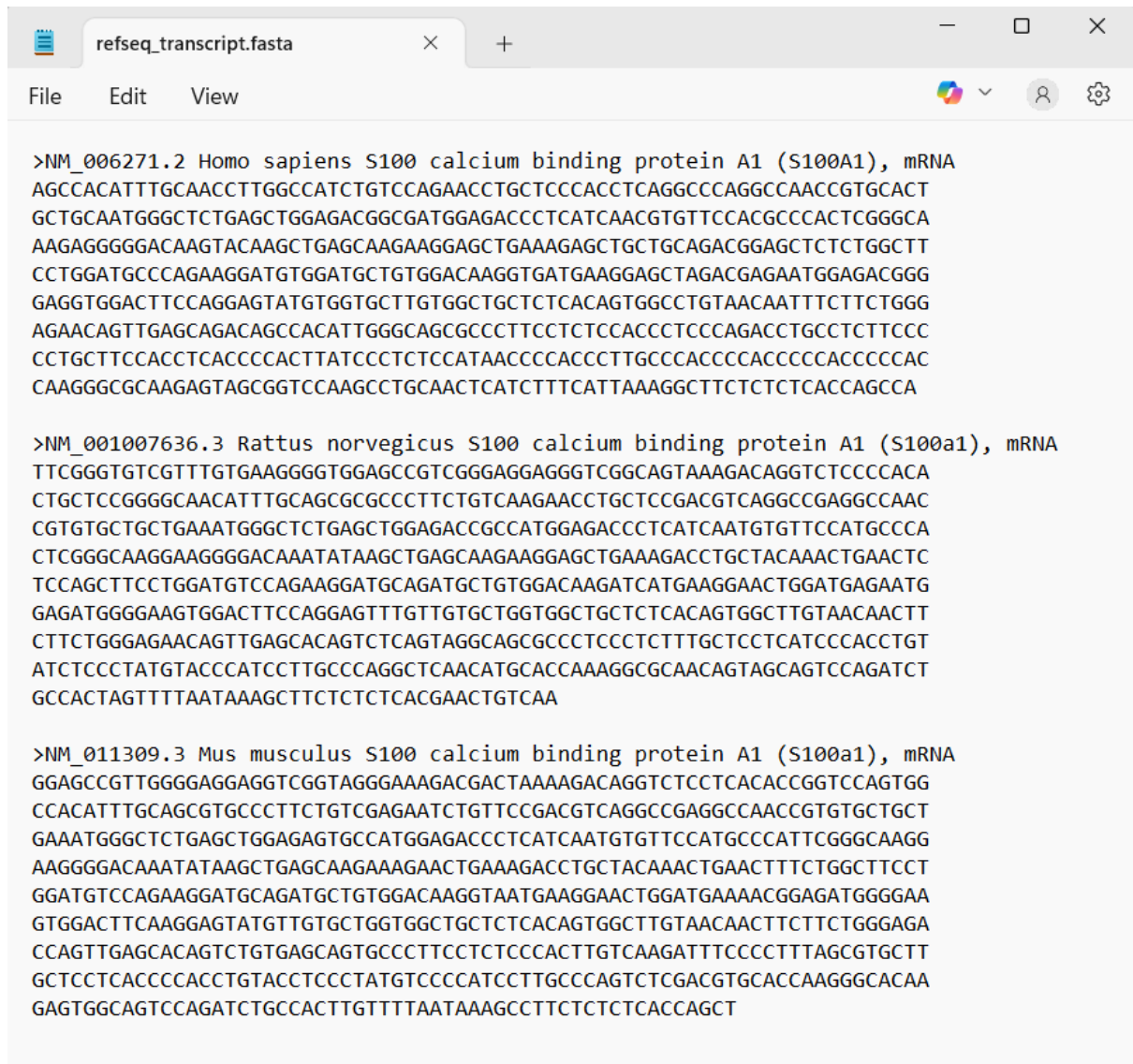
☒ all sequences per gene

Download

Feedback

FASTA :





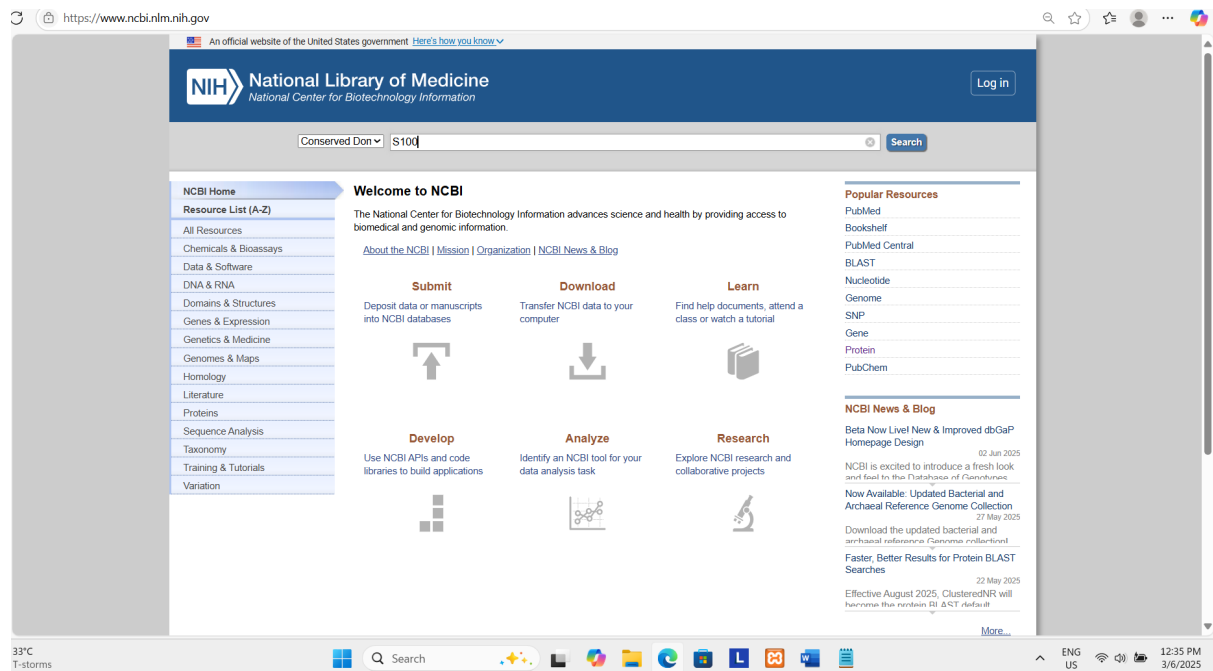
```
>NM_006271.2 Homo sapiens S100 calcium binding protein A1 (S100A1), mRNA
AGCCACATTTGCAACCTTGCCATCTGTCCAGAACCTGCTCCACCTCAGGCCAGGCCAACCGTGCACT
GCTGCAATGGGCTCTGAGCTGGAGACGGCGATGGAGACCCTCATCAACGTGTTCCACGCCACTCGGGCA
AAGAGGGGGACAAGTACAAGCTGAGCAAGAAGGAGCTGAAAGAGCTGCTGCAGACGGAGCTCTTGCTT
CCTGGATGCCCCAGAAGGATGTGGATGCTGTGGACAAGGTGATGAAGGAGCTAGACGAGAATGGAGACGG
GAGGTGGACTTCCAGGAGTATGTGGTGCTTGTGGCTGCTCTCACAGTGGCCTGTAACAATTTCTTCTGG
AGAACAGTTGAGCAGACAGCCACATTGGGCAGCGCCCTTCTCTCCACCCTCCAGACCTGCCTCTTCCC
CCTGCTTCCACCTCACCCACTTATCCCTCTCCATAACCCACCCTTGCCACCCACCCCAACCCAC
CAAGGGCGCAAGAGTAGCGGTCCAAGCCTGCAACTCATCTTTCATTAAAGGCTTCTCTCACCAGCCA

>NM_001007636.3 Rattus norvegicus S100 calcium binding protein A1 (S100a1), mRNA
TTCGGGTGTCGTTTGTGAAGGGGTGGAGCCGTCGGGAGGAGGGTCGGCAGTAAAGACAGGTCTCCCCACA
CTGCTCCGGGGCAACATTTGCAGCGCGCCCTTCTGTCAAGAACCTGCTCCGACGTCAGGCCGAGGCCAAC
CGTGTGCTGCTGAAATGGGCTCTGAGCTGGAGACCGCCATGGAGACCCTCATCAATGTGTTCCATGCCCA
CTCGGGCAAGGAAGGGGACAAATATAAGCTGAGCAAGAAGGAGCTGAAAGACCTGCTACAACTGAACTC
TCCAGCTTCTTGATGTCCAGAAGGATGCAGATGCTGTGGACAAGATCATGAAGGAAGTGGATGAGAATG
GAGATGGGGAAGTGGACTTCCAGGAGTTTGTGTGCTGGTGGCTGCTCTCACAGTGGCTTGTAACAATT
CTTCTGGGAGAACAGTTGAGCACAGTCTCAGTAGGCAGCGCCCTCCCTCTTTGCTCCTCATCCACCTGT
ATCTCCCTATGTACCCATCCTTGCCCAGGCTCAACATGCACCAAAGGCGCAACAGTAGCAGTCCAGATCT
GCCACTAGTTTTAATAAAGCTTCTCTCTCACGAAGTGTCAA

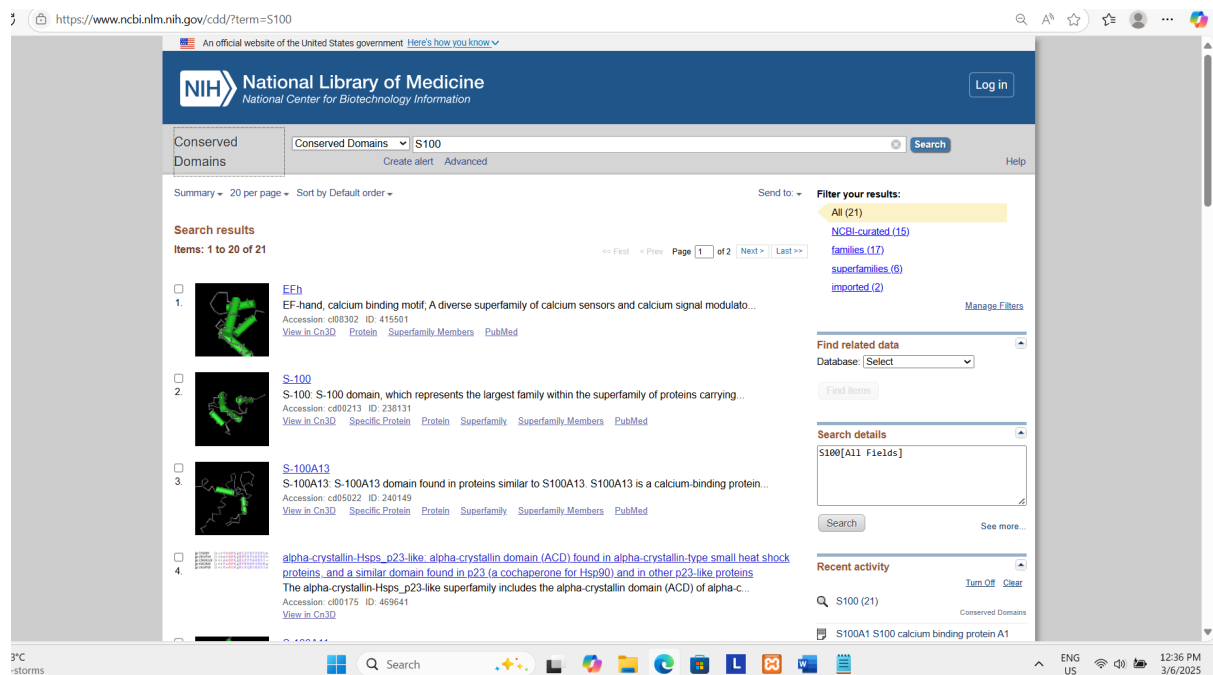
>NM_011309.3 Mus musculus S100 calcium binding protein A1 (S100a1), mRNA
GGAGCCGTTGGGGAGGAGGTCGGTAGGGAAAGACGACTAAAAGACAGGTCTCTCACACCGGTCCAGTGG
CCACATTTGCAGCGTGCCTTCTGTCTGAGAATCTGTTCCGACGTCAGGCCGAGGCCAACCGTGTGCTGCT
GAAATGGGCTCTGAGCTGGAGAGTGCCATGGAGACCCTCATCAATGTGTTCCATGCCCATTCGGGCAAGG
AAGGGGACAAATATAAGCTGAGCAAGAAAGAACTGAAAGACCTGCTACAACTGAACTTCTGGCTTCT
GGATGTCCAGAAGGATGCAGATGCTGTGGACAAGGTAATGAAGGAAGTGGATGAAAACGGAGATGGGGAA
GTGGACTTCAAGGAGTATGTTGTGCTGGTGGCTGCTCTCACAGTGGCTTGTAACAATTCTTCTGGGAGA
CCAGTTGAGCACAGTCTGTGAGCAGTGCCCTTCTCTCCCACTTGTCAGATTTCCCTTTAGCGTGCTT
GCTCCTCACCCACCTGTACCTCCCTATGTCCCATCCTTGCCCAGTCTCGACGTGCACCAAGGGCACAA
GAGTGGCAGTCCAGATCTGCCACTTGTTTTAATAAAGCCTTCTCTCTCACCAGCT
```

## SECOND APPROACH:

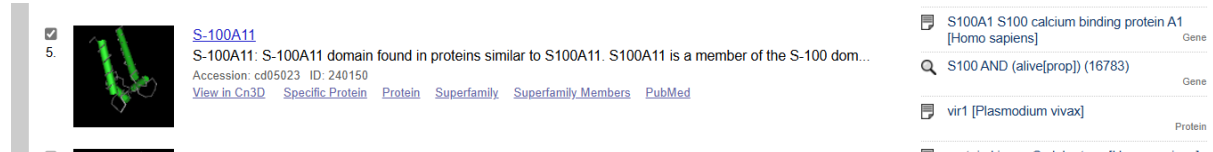
At homepage of NCBI select conserved domain and search for S100



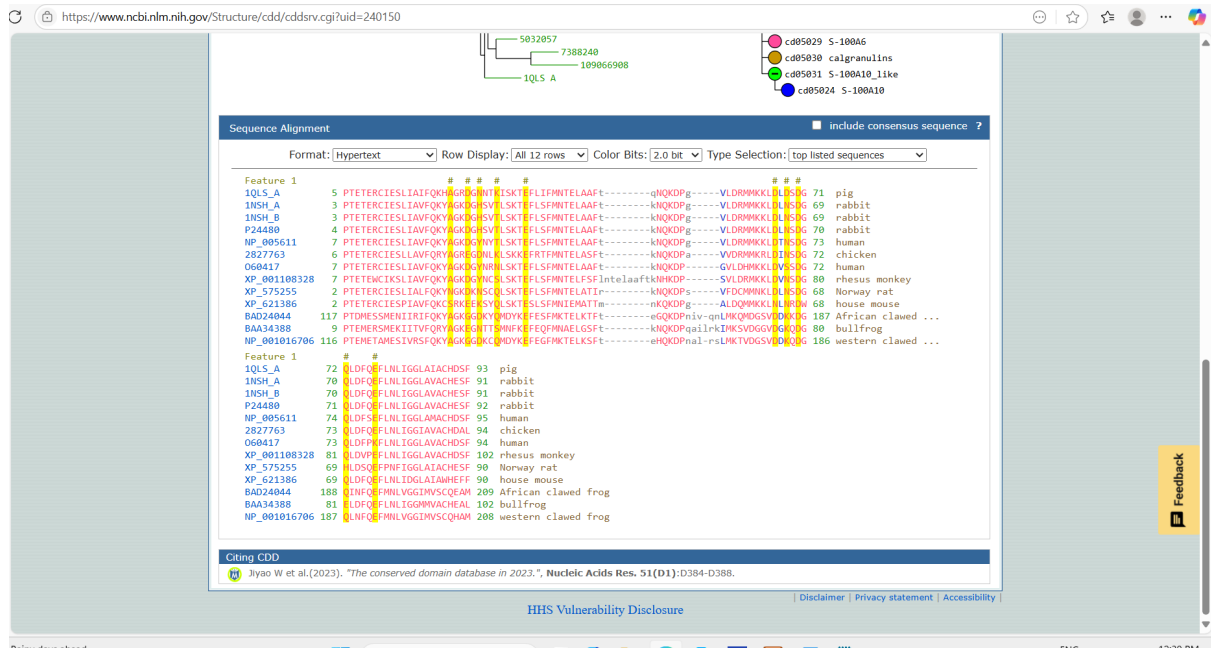
Here the results :



The results include S-100A11:



That entry includes structure information and also a multiple sequence alignment that we can reformat as a text file or multiple FASTA file to save and study further.



### THIRD APPROACH:

To start with third approach, we perform BLASTP using ferritin (NP\_000137) as a query, restricting the output to the RefSeq database.

Enter Query Sequence

Enter accession number(s), gis(s), or FASTA sequence(s)

Query subrange: From  To

Or upload file:  No file chosen

Job Title:

☐ Align two or more sequences

Choose Search Set

Database: ☒ Standard databases (or etc) ☐ ClusteredNR database ☐ Reference proteins (refseq protein)

Organism:

Exclude: ☐ Models (OMXP) ☐ Non-redundant RefSeq proteins (NRP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm: ☒ blastp (protein-protein BLAST) ☐ PSI-BLAST (Position-Specific Iterated BLAST) ☐ PHI-BLAST (Pattern Hit Initiated BLAST) ☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Search database refseq\_protein using Blastp (protein-protein BLAST)

☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with a sign

Algorithm parameters

Here are the results, and select 10 of the hits (by checking the boxes).

https://blast.ncbi.nlm.nih.gov/Blast.cgi

NIH National Library of Medicine  
National Center for Biotechnology Information

BLAST® » blastp suite » results for RID-3WZFS6N013

**Important update**  
Effective August 2025, the **ClusteredNR** database will become the default Protein BLAST database. Learn more about ClusteredNR

[Edit Search](#) [Save Search](#) [Search Summary](#) [How to read this report?](#) [BLAST Help Videos](#) [Back to Traditional Results Page](#)

Job Title: NP\_000137:ferritin light chain [Homo sapiens]  
RID: 3WZFS6N013 Search expires on 06-04 12:45 pm [Download All](#)  
Program: BLASTP [Citation](#)  
Database: refseq\_protein [See details](#)  
Query ID: NP\_000137.2  
Description: ferritin light chain [Homo sapiens]  
Molecule type: amino acid  
Query Length: 175  
Other reports: [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

**Filter Results**  
Organism: only top 20 will appear ☐ exclude  
Type common name, binomial, taxid or group name  
[Add organism](#)  
Percent Identity:  to  E value:  to  Query Coverage:  to   
[Filter](#) [Reset](#)

**Sequences producing significant alignments** Download Select columns Show 100

| Description                                                                                             | Scientific Name            | Max Score | Total Score | Query Cover | E value | Per. Ident | Acc. Len | Accession      |
|---------------------------------------------------------------------------------------------------------|----------------------------|-----------|-------------|-------------|---------|------------|----------|----------------|
| <input checked="" type="checkbox"/> ferritin light chain [Homo sapiens]                                 | Homo sapiens               | 360       | 360         | 100%        | 2e-125  | 100.00%    | 175      | NP_000137.2    |
| <input checked="" type="checkbox"/> ferritin light chain [Pan paniscus]                                 | Pan paniscus               | 357       | 357         | 100%        | 2e-124  | 99.43%     | 175      | XP_000959541.3 |
| <input checked="" type="checkbox"/> ferritin light chain [Pan troglodytes]                              | Pan troglodytes            | 357       | 357         | 100%        | 2e-124  | 98.86%     | 175      | NP_001128650.1 |
| <input checked="" type="checkbox"/> ferritin light chain-like [Pan troglodytes]                         | Pan troglodytes            | 358       | 358         | 100%        | 1e-123  | 99.43%     | 244      | XP_003672238.1 |
| <input checked="" type="checkbox"/> ferritin light chain [Rhinopithecus roosevelti]                     | Rhinopithecus roosevelti   | 353       | 353         | 100%        | 1e-122  | 97.71%     | 175      | XP_010369818.1 |
| <input checked="" type="checkbox"/> ferritin light chain [Macaca mulatta]                               | Macaca mulatta             | 352       | 352         | 100%        | 2e-122  | 97.14%     | 175      | NP_001248136.1 |
| <input checked="" type="checkbox"/> ferritin light chain [Ptilinopus leucosceles]                       | Ptilinopus leucosceles     | 351       | 351         | 100%        | 5e-122  | 97.14%     | 175      | XP_023038337.1 |
| <input checked="" type="checkbox"/> ferritin light chain-like [Trichobates moloch]                      | Trichobates moloch         | 351       | 351         | 100%        | 6e-122  | 97.71%     | 175      | XP_032610358.1 |
| <input checked="" type="checkbox"/> PREDICTED: ferritin light chain isoform X2 [Mandillus leucochaesus] | Mandillus leucochaesus     | 350       | 350         | 100%        | 6e-122  | 96.57%     | 175      | XP_011834728.1 |
| <input checked="" type="checkbox"/> ferritin light chain-like [Macaca thibetana thibetana]              | Macaca thibetana thibetana | 352       | 352         | 100%        | 2e-121  | 97.14%     | 226      | XP_056837471.1 |

33°C Mostly cloudy

A download menu appears, choose download FASTA (completesequences)

https://blast.ncbi.nlm.nih.gov/Blast.cgi

Job Title: NP\_000137:ferritin light chain [Homo sapiens]  
RID: 3WZFS6N013 Search expires on 06-04 12:45 pm [Download All](#)  
Program: BLASTP [Citation](#)  
Database: refseq\_protein [See details](#)  
Query ID: NP\_000137.2  
Description: ferritin light chain [Homo sapiens]  
Molecule type: amino acid  
Query Length: 175  
Other reports: [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

**Filter Results**  
Organism: only top 20 will appear ☐ exclude  
Type common name, binomial, taxid or group name  
[Add organism](#)  
Percent Identity:  to  E value:  to  Query Coverage:  to   
[Filter](#) [Reset](#)

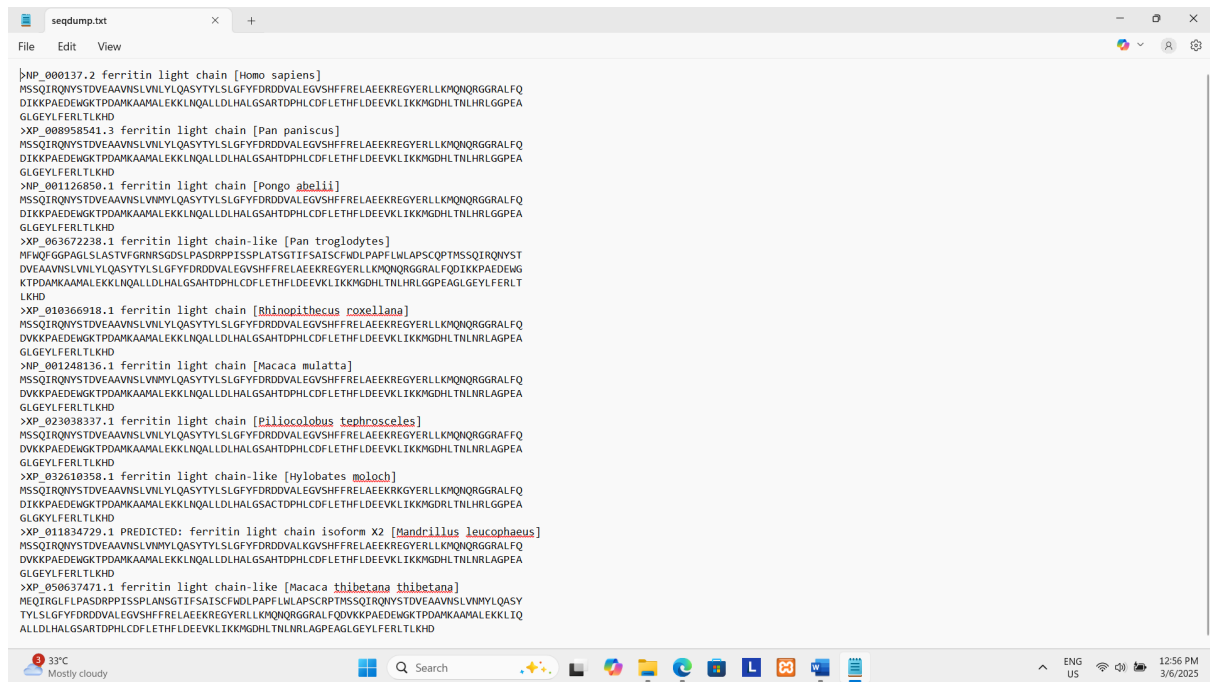
**Sequences producing significant alignments** Download Select columns Show 100

| Description                                                                                             | Scientific Name            | Max Score | Total Score | Query Cover | E value | Per. Ident | Acc. Len | Accession      |
|---------------------------------------------------------------------------------------------------------|----------------------------|-----------|-------------|-------------|---------|------------|----------|----------------|
| <input checked="" type="checkbox"/> ferritin light chain [Homo sapiens]                                 | Homo sapiens               | 360       | 360         | 100%        | 2e-125  | 100.00%    | 175      | NP_000137.2    |
| <input checked="" type="checkbox"/> ferritin light chain [Pan paniscus]                                 | Pan paniscus               | 357       | 357         | 100%        | 2e-124  | 99.43%     | 175      | XP_000959541.3 |
| <input checked="" type="checkbox"/> ferritin light chain [Pan troglodytes]                              | Pan troglodytes            | 357       | 357         | 100%        | 2e-124  | 98.86%     | 175      | NP_001128650.1 |
| <input checked="" type="checkbox"/> ferritin light chain-like [Pan troglodytes]                         | Pan troglodytes            | 358       | 358         | 100%        | 1e-123  | 99.43%     | 244      | XP_003672238.1 |
| <input checked="" type="checkbox"/> ferritin light chain [Rhinopithecus roosevelti]                     | Rhinopithecus roosevelti   | 353       | 353         | 100%        | 1e-122  | 97.71%     | 175      | XP_010369818.1 |
| <input checked="" type="checkbox"/> ferritin light chain [Macaca mulatta]                               | Macaca mulatta             | 352       | 352         | 100%        | 2e-122  | 97.14%     | 175      | NP_001248136.1 |
| <input checked="" type="checkbox"/> ferritin light chain [Ptilinopus leucosceles]                       | Ptilinopus leucosceles     | 351       | 351         | 100%        | 5e-122  | 97.14%     | 175      | XP_023038337.1 |
| <input checked="" type="checkbox"/> ferritin light chain-like [Trichobates moloch]                      | Trichobates moloch         | 351       | 351         | 100%        | 6e-122  | 97.71%     | 175      | XP_032610358.1 |
| <input checked="" type="checkbox"/> PREDICTED: ferritin light chain isoform X2 [Mandillus leucochaesus] | Mandillus leucochaesus     | 350       | 350         | 100%        | 6e-122  | 96.57%     | 175      | XP_011834728.1 |
| <input checked="" type="checkbox"/> ferritin light chain-like [Macaca thibetana thibetana]              | Macaca thibetana thibetana | 352       | 352         | 100%        | 2e-121  | 97.14%     | 226      | XP_056837471.1 |
| <input type="checkbox"/> PREDICTED: ferritin light chain isoform X2 [Colinus arropensis pallasi]        | Colinus arropensis pallasi | 346       | 346         | 100%        | 4e-120  | 97.14%     | 174      | XP_011801814.1 |
| <input type="checkbox"/> ferritin light chain isoform X2 [Theropithecus veldi]                          | Theropithecus veldi        | 346       | 346         | 100%        | 6e-120  | 96.57%     | 174      | XP_025259890.1 |
| <input type="checkbox"/> PREDICTED: ferritin light chain isoform X1 [Mandillus leucochaesus]            | Mandillus leucochaesus     | 344       | 344         | 100%        | 3e-119  | 94.92%     | 177      | XP_011834728.1 |
| <input type="checkbox"/> PREDICTED: ferritin light chain isoform X3 [Mandillus leucochaesus]            | Mandillus leucochaesus     | 344       | 344         | 100%        | 3e-119  | 96.00%     | 174      | XP_011834720.1 |
| <input type="checkbox"/> PREDICTED: ferritin light chain isoform X4 [Mandillus leucochaesus]            | Mandillus leucochaesus     | 342       | 342         | 100%        | 3e-118  | 96.00%     | 174      | XP_011834731.1 |
| <input type="checkbox"/> ferritin light chain-like [Pongo pygmaeus]                                     | Pongo pygmaeus             | 341       | 341         | 100%        | 6e-118  | 94.96%     | 175      | XP_063515096.1 |
| <input type="checkbox"/> ferritin light chain [Aotus nasutus]                                           | Aotus nasutus              | 340       | 340         | 100%        | 1e-117  | 93.71%     | 175      | XP_012315624.1 |
| <input type="checkbox"/> ferritin light chain [Gorilla gorilla gorilla]                                 | Gorilla gorilla gorilla    | 340       | 340         | 100%        | 1e-117  | 94.96%     | 175      | XP_018872821.1 |
| <input type="checkbox"/> ferritin light chain-like [Pan troglodytes]                                    | Pan troglodytes            | 340       | 340         | 100%        | 1e-117  | 94.96%     | 175      | XP_063659346.1 |
| <input type="checkbox"/> ferritin light chain [Callithrix jacchus]                                      | Callithrix jacchus         | 339       | 339         | 100%        | 3e-117  | 93.14%     | 175      | XP_008866896.2 |

https://blast.ncbi.nlm.nih.gov/Blast.cgi

33°C Mostly cloudy

Upon downloading you obtain a text file containing sequences, here in the FASTA format.



Here are 10 ferritin proteins:

>NP\_000137.2 ferritin light chain [Homo sapiens]

```
MSSQIRQNYSTDVEAAVNSLVNLYQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQ
DIKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEA
GLGEYLFERLTLKHD
```

>XP\_008958541.3 ferritin light chain [Pan paniscus]

```
MSSQIRQNYSTDVEAAVNSLVNLYQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQ
DIKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEA
GLGEYLFERLTLKHD
```

>NP\_001126850.1 ferritin light chain [Pongo abelii]

```
MSSQIRQNYSTDVEAAVNSLVNMYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQ
DIKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEA
GLGEYLFERLTLKHD
```

>XP\_063672238.1 ferritin light chain-like [Pan troglodytes]

```
MFWQFGGPAGLSLASTVFGNRSGDSLPAADRPPISPLATSGTIFSAISCFWDLPAFLWLAPSCQPTMSSQIRQNYST
DVEAAVNSLVNLYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQDIKKPAEDEWG
KTPDAMKAAMALEKKLNQALLDLHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGEYLFERLT
LKHD
```

>XP\_010366918.1 ferritin light chain [Rhinopithecus roxellana]

MSSQIRQNYSTDVEAAVNSLVNLYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQ  
DVKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNRLAGPEA  
GLGEYLFERLTLKHD

>NP\_001248136.1 ferritin light chain [Macaca mulatta]

MSSQIRQNYSTDVEAAVNSLVNMYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQ  
DVKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNRLAGPEA  
GLGEYLFERLTLKHD

>XP\_023038337.1 ferritin light chain [Piliocolobus tephrosceles]

MSSQIRQNYSTDVEAAVNSLVNLYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRAFFQ  
DVKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNRLAGPEA  
GLGEYLFERLTLKHD

>XP\_032610358.1 ferritin light chain-like [Hylobates moloch]

MSSQIRQNYSTDVEAAVNSLVNLYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKRKGYERLLKMQNQRGGRALFQ  
DIKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSACTDPHLCDFLETHFLDEEVKLIKMGDRLTNLHRLGGPEA  
GLGKYLFERLTLKHD

>XP\_011834729.1 PREDICTED: ferritin light chain isoform X2 [Mandrillus leucophaeus]

MSSQIRQNYSTDVEAAVNSLVNMYLQASYTYLSLGFYFDRDDVALKGVSFFRELAEEKREGYERLLKMQNQRGGRALFQ  
DVKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNRLAGPEA  
GLGEYLFERLTLKHD

>XP\_050637471.1 ferritin light chain-like [Macaca thibetana thibetana]

MEQIRGLFLPASDRPPISSPLANSGETIFSACFWDLAPFLWLAPSCRPTMSSQIRQNYSTDVEAAVNSLVNMYLQASY  
TYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQDVKKPAEDEWGKTPDAMKAAMALEKKLIQ  
ALLDLHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNRLAGPEAGLGEYLFERLTLKHD

It can be helpful to simplify the header lines, as follows; these are ferritin light chains from five species.

>Human NP\_000137.2

MSSQIRQNYSTDVEAAVNSLVNLYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQ  
DIKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEA  
GLGEYLFERLTLKHD

> Pan paniscus XP\_008958541.3

MSSQIRQNYSTDVEAAVNSLVNLYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQ  
DIKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEA  
GLGEYLFERLTLKHD

> *Pongo abelii* NP\_001126850.1

MSSQIRQNYSTDVEAAVNSLVNMYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQ  
DIKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEA  
GLGEYLFERLTLKHD

> *Pan troglodytes* XP\_063672238.1

MFWQFGGPAGLSLASTVFGRNRSGDSLPAADRPPISPLATSGTIFSAISCFWDLPAFLWLAPSCOPTMSSQIRQNYST  
DVEAAVNSLVNLYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQDIKKPAEDEWG  
KTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGEYLFERLT  
LKHD

> *Rhinopithecus roxellana* XP\_010366918.1

MSSQIRQNYSTDVEAAVNSLVNLYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQ  
DVKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNRLAGPEA  
GLGEYLFERLTLKHD

> *Macaca mulatta* NP\_001248136.1

MSSQIRQNYSTDVEAAVNSLVNMYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQ  
DVKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNRLAGPEA  
GLGEYLFERLTLKHD

> *Ptilocolobus tephrosceles* XP\_023038337.1

MSSQIRQNYSTDVEAAVNSLVNLYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRAFFQ  
DVKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNRLAGPEA  
GLGEYLFERLTLKHD

> *Hylobates moloch* XP\_032610358.1

MSSQIRQNYSTDVEAAVNSLVNLYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKRKGYERLLKMQNQRGGRALFQ  
DIKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSACTDPHLCDFLETHFLDEEVKLIKMGDRLTNLHRLGGPEA  
GLGKYLFERLTLKHD

> *Mandrillus leucophaeus* XP\_011834729.1

MSSQIRQNYSTDVEAAVNSLVNMYLQASYTYLSLGFYFDRDDVALKGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQ  
DVKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNRLAGPEA

GLGEYLFERLTLKHD

> *Macaca thibetana thibetana* XP\_050637471.1

MEQIRGLFLPASDRPPISSPLANS GTIFSAISCFWDL PAPFLWLAPSCRPTMSSQIRQNYSTDVEAAVNSLVNMYLQASY

TYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQDVKKPAEDEWGKTPDAMKAAMALEKKLIQ

ALLDLHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNLRLAGPEAGLGEYLFERLTLKHD

2) Using the FASTA-formatted sequences from problem (1), perform multiple sequence alignments using programs available at the European Bioinformatics Institute: MAFFT, Muscle, and T-Coffee.

- i. Save and compare each result. How do they differ? How can you assess which is likely to be the most accurate?
- ii. When applicable, try adjusting the parameters such as the scoring matrices, gap opening and extension penalties, or number of iterations to see the effects on the alignments.

#### (2) SOLUTIONS/COMMENTS:

If you choose a set of sequences from a HomoloGene entry it is possible (even likely) that they will all be so closely related that all multiple alignment programs give equal (or closely similar) results. It is more interesting to find sequences that are less related. One approach is to use the Conserved Domain Database (as shown in the previous problem, bottom image) and select the most diverse members of a set.

Search for trypsin :



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Rainy days ahead  
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https://www.ncbi.nlm.nih.gov/cdd/?term=trypsin

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☐ **beta-trefoil\_STI: soybean trypsin inhibitor (STI)-like domain, beta-trefoil fold**  
The STI-like domain is found in the soybean trypsin inhibitor (Kunitz) family of protease inhibitors...  
Accession: c11496 ID: 472192  
[View in Cn3D](#)

☐ **Peptidases\_S8\_S53: Peptidase domain in the S8 and S53 families**  
Members of the peptidases S8 (subtilisin and kexin) and S53 (sedolisin) family include endopeptidase...  
Accession: c10459 ID: 415849  
[View in Cn3D](#) [Protein](#) [Superfamily Members](#) [PubMed](#)

☐ **Trypsin, Trypsin**  
Accession: pfam00089 ID: 459667  
[View in Cn3D](#)

☐ **beta-trefoil\_STI\_LITI-like: soybean trypsin inhibitor (STI)-like domain, beta-trefoil fold, found in Leucaena leucocephala trypsin inhibitor (LITI) and similar proteins**  
LITI is a Kunitz-type trypsin inhibitor that inhibits trypsin, plasmin, human plasma kallikrein, chy...  
Accession: cd23365 ID: 467390  
[View in Cn3D](#)

☐ **beta-trefoil\_STI\_AITPI-like: soybean trypsin inhibitor (STI)-like domain, beta-trefoil fold, found in Arabidopsis thaliana trypsin protease inhibitor (AITPI) and similar proteins**  
AITPI, also called kunitz trypsin inhibitor 4 (AITK14), or Kunitz trypsin inhibitor 1 (AITK11), exhi...  
Accession: cd23366 ID: 467391  
[View in Cn3D](#)

☐ **beta-trefoil\_STI\_SKTI: soybean trypsin inhibitor (STI)-like domain, beta-trefoil fold, found in the soybean Kunitz trypsin inhibitor (SKTI) subfamily**  
SKTI is extracted from soybean (Glycine max L.) seeds. It shows inhibition of trypsin and possesses ...  
Accession: cd23363 ID: 467388  
[View in Cn3D](#)

☐ **beta-trefoil\_STI\_COTI: soybean trypsin inhibitor (STI)-like domain, beta-trefoil fold, found in Senna obtusifolia trypsin inhibitor 1 (COTI) and similar proteins**

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**Recent activity**

trypsin (208)  
Conserved Domains

S100 (21)  
Conserved Domains

S100A1 S100 calcium binding protein A1 [Homo sapiens]  
Gene

S100 AND (alive[prop]) (16783)  
Gene

vir1 [Plasmodium vivax]  
Protein

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Search

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Trypsin results:

https://www.ncbi.nlm.nih.gov/Structure/cdd/cddsrv.cgi?uid=459667

## Conserved Protein Domain Family

### Trypsin

pfam00089: Trypsin (this model, PSSM-Id:459667 is obsolete and has been replaced by 513138) [Download alignment ?](#)

Trypsin

**Links**

- Source: pfam
- Taxonomy: cellular organisms
- PubMed: 2 links
- Protein: Representatives
- Specific Protein
- Related Protein
- Related Structure
- Architectures
- Superfamily: c47494

**PubMed References**

- The three-dimensional structure of Asn102 mutant of trypsin: role of Asp102 in serine protease catalysis. *Science* 1987 Aug 21; 237(4817):905-9
- Families of serine peptidases. *Methods Enzymol* 1994 ; 244:19-61

**Sequence trees and CD hierarchy information associated with this CD can not be displayed.**

**This model has been replaced with 513138.**

**Sequence Alignment** ☐ include consensus sequence ?

Format: [Hypertext](#) Row Display: [up to 10](#) Color Bits: [2.0 bit](#) Type Selection: [the most diverse members](#)

|        |     |                                                                                 |     |                                                     |
|--------|-----|---------------------------------------------------------------------------------|-----|-----------------------------------------------------|
| 20K5_A | 466 | VIEIRKGTQYIKapWQAKISVIRpskgHeSCGAVVSEY---FVLTAHCF---VDD---KE-HSIVKSVGG-KRDL     | 533 | human                                               |
| Q2YMK6 | 125 | PIPGHERPVAGC-----S-----GFVISEDg-YVVTNNHVSD-----GdaYTVLDDGTeLDAKL                | 174 | Brucella abortus 2308                               |
| P26982 | 97  | QGGNGNGGQQ---QKFMALGS-----GVIIIDAAKgyVVTNNHVSDn---a5v---IK---VQLSDGn---KFDA     | 152 | Salmonella enterica s...                            |
| Q44596 | 85  | SGFGASKPRIQQ-----SLGS-----GVIVDRSG-IYVNNHVIKda-deIKVa-LSO---GREFFES---RILL      | 139 | Brucella abortus                                    |
| P39668 | 99  | DIWESGEAGSG---SGVITYKN---DHSA---YVVTNNHVIEg-----a5Q---IEISLK-----DG             | 144 | Bacillus subtilis sub...                            |
| 3LGV_C | 47  | YHRIKNTSHHQ---LEIRTl-gS---GVVIDBg-YVVTNNHVIMd---IKQD-----96                     | 96  | Escherichia coli K-12                               |
| P44947 | 37  | ITFSKNAVRIASp-AVVNVYHRS---FSSASINDND-Q-LQVNLGSGvimsKdGylITkHLITQNAQI--VVAL      | 104 | Haemophilus influenza...                            |
| 066439 | 59  | PGGGPQRQHAD-----ALGS-----GFVISADg-YVVTNNHVIKge-dGNTv-aDE---ISIEFANGKLLDA        | 115 | Rhodobacter capsulatus                              |
| AS1TX7 | 37  | ENNVTKIQDTNI fpyTGVVAFKS-----ATGFVVGKIN--TILTNKHVSKn---KVGDR-iTAHPSDKNGG---IY51 | 102 | Staphylococcus aureus...                            |
| AS1TX6 | 38  | KIVTQKQDTNI fpyYGVVSKFD-----ATGVVIGKIN--TILTNKHVSkd---YKV---GD-RITAHPMWdKgNGI   | 99  | Staphylococcus aureus...                            |
| 20K5_A | 534 | FEevLFHPHRYNGKkeagipefydYVALIRKRNK-----LYGQGTIPICPtegttRaiRlPPTTCQQQE           | 685 | human                                               |
| Q2YMK6 | 175 | IG-----ADPRT---DLAVLKTNP-----KRFV-YVAFGD---Dn---KVRVSGWNVAV                     | 215 | Brucella abortus 2308                               |
| P26982 | 153 | KV---VG---KDPK-----DIALVKTQNP-----KINLTATKLADSD-----ALRVGYTVAT                  | 194 | Salmonella enterica s...                            |
| Q44596 | 140 | RD-----ET-----DLAVLKTIEAK-----QFPVLALGNSD-----EVEVGDLVAT                        | 177 | Brucella abortus                                    |
| P39668 | 145 | SR---VSADLVGSDQI-----DLAVLKRKSD-----KIKAVADFGNSD-----KVGSGEPVAT                 | 191 | Bacillus subtilis sub...                            |
| 3LGV_C | 97  | Rv---FEALVGSQSI-----DLAVLKTINAT-----GLPTTIPINARR-----VPHGQVVLAT                 | 143 | Escherichia coli K-12                               |
| P44947 | 105 | QNgIIFEALVGSQSI-----DLAVLKTIRAD-----NLSTIPQNSAR-----QAHVGQVVLAT                 | 153 | Haemophilus influenza...                            |
| 066439 | 116 | KIV---GTDPKT-----DIALLVKESK-----EPLPFVSGDSOT-----HREGQVNMAM                     | 157 | Rhodobacter capsulatus                              |
| AS1TX7 | 103 | KKI---IN---YPGK-----DVSIVQEERaiergpkGFNDNVPFKYAA---Ga---KAGERIKVIGY             | 157 | Staphylococcus aureus...                            |
| AS1TX6 | 100 | YKI---KSTSDYPPGD-----DISVWNIEEQaiergpkGFNERNVQAFYAA---Da---KVDDIKVIGY           | 157 | Staphylococcus aureus...                            |
| 20K5_A | 606 | ELIpaGDKTAlfveeeekLLTRKEVTVINMGKGSCEERdayAPGYdkvkdsVTPRFLCTGvsspydntICRGSG      | 685 | human                                               |
| Q2YMK6 | 216 | GN---PFGLG-----GTVTSIGVSALGRSD-IGAGP-----YDFITQIDA-----AVNRGNSG                 | 258 | Brucella abortus 2308                               |
| P26982 | 195 | GN---PFGLG-----ETVTSIGVSALGRSLINVEN-----YDFITQIDA-----AINRGNSG                  | 238 | Salmonella enterica s...                            |
| Q44596 | 178 | GN---PFGVG-----QTVTSIGVSASRTQVTSDF-----DEFITQIDA-----AINPGNSG                   | 221 | Brucella abortus                                    |
| P39668 | 192 | GhIglEFAG-----SVTQGVISGTERAIPVDSNg-dGQpd-----WNAEVLQIDA-----AINPGNSG            | 244 | Bacillus subtilis sub...                            |
| 3LGV_C | 144 | GN---PYNLG-----QTITGGLISATGRILHPT-----GR-----QNFITQIDA-----SINRGNSG             | 187 | Escherichia coli K-12                               |
| P44947 | 154 | GN---PYNLG-----QSVSGTISATGRNAGDSVg-----RQNFITQIDA-----SINRGNSG                  | 198 | Haemophilus influenza...                            |
| 066439 | 158 | GhIp---LG-----QGFSTSGVLVSA---RNRe-LSGTY-----DCCIQTIDA-----SINRGNSG              | 199 | Rhodobacter capsulatus                              |
| AS1TX7 | 158 | PHI---PYKNC-----YVLYESTGPNVSE-GSSIV-----YSAH---TES-----GNSG                     | 194 | Staphylococcus aureus...                            |
| AS1TX6 | 158 | Plp-pQISFK-----QFE-----STG-TIKR-----IDKRELNFDA-----YIEPGNSG                     | 194 | Staphylococcus aureus...                            |
| 20K5_A | 686 | GPLTHKRSrFIQGVISMGvDCKNQKqkvPAHA-RDFHInlfQVLPLH                                 | 737 | human                                               |
| Q2YMK6 | 259 | GPADLSG---EVIGINTAI---FSPSGGSVg---IAFA-IPSS---TAKQV                             | 298 | Brucella abortus 2308                               |
| P26982 | 239 | GALVNLG---ELIGINTAI---LAPOGGIlg---IGA-IPSN---MVKNL                              | 278 | Salmonella enterica subsp. enterica serovar Typh... |
| Q44596 | 222 | GALLIDRG---RLIGINTAI---YSRSGGSVg---IGA-IPSN---MVRVAV                            | 261 | Brucella abortus                                    |
| P39668 | 245 | GALLIDRG---KVTIGNSP---IAEAVG---IGLS-IPSN---LVIPDI                               | 283 | Bacillus subtilis subsp. subtilis str. 168          |
| 3LGV_C | 188 | GALVNSLG---ELMGINTLSF-DKSNIDGfne-gIGA-IPFG---LATK                               | 229 | Escherichia coli K-12                               |
| P44947 | 199 | GALINSAG---ELVGISTLSIgKTANIEAte---LNFA-IPID---IANDVL                            | 240 | Haemophilus influenzae Rd KW20                      |
| 066439 | 200 | GPLFSMDG---KVGIVNTAI---LSPNG---STG-IGFS---MASNV                                 | 235 | Rhodobacter capsulatus                              |
| AS1TX7 | 195 | SPVLNSNI---ELVGIFHAS---DKNIDDIIn---aGVYfTPEI---KKF                              | 234 | Staphylococcus aureus subsp. aureus JH9             |
| AS1TX6 | 195 | SPVLNSNI---ELVGIVYGGI-GKIGSE---YNGA-VYfTp---QIKDF                               | 233 | Staphylococcus aureus subsp. aureus JH9             |

gh UV  
ow

https://www.ncbi.nlm.nih.gov/Structure/cdd/cddsrv.cgi?uid=459667

**Sequence Alignment** ☐ include consensus sequence ?

Format: [Hypertext](#) Row Display: [up to 10](#) Color Bits: [2.0 bit](#) Type Selection: [the most diverse members](#)

|        |     |                                                                                 |     |                                                     |
|--------|-----|---------------------------------------------------------------------------------|-----|-----------------------------------------------------|
| 20K5_A | 466 | VIEIRKGTQYIKapWQAKISVIRpskgHeSCGAVVSEY---FVLTAHCF---VDD---KE-HSIVKSVGG-KRDL     | 533 | human                                               |
| Q2YMK6 | 125 | PIPGHERPVAGC-----S-----GFVISEDg-YVVTNNHVSD-----GdaYTVLDDGTeLDAKL                | 174 | Brucella abortus 2308                               |
| P26982 | 97  | QGGNGNGGQQ---QKFMALGS-----GVIIIDAAKgyVVTNNHVSDn---a5v---IK---VQLSDGn---KFDA     | 152 | Salmonella enterica s...                            |
| Q44596 | 85  | SGFGASKPRIQQ-----SLGS-----GVIVDRSG-IYVNNHVIKda-deIKVa-LSO---GREFFES---RILL      | 139 | Brucella abortus                                    |
| P39668 | 99  | DIWESGEAGSG---SGVITYKN---DHSA---YVVTNNHVIEg-----a5Q---IEISLK-----DG             | 144 | Bacillus subtilis sub...                            |
| 3LGV_C | 47  | YHRIKNTSHHQ---LEIRTl-gS---GVVIDBg-YVVTNNHVIMd---IKQD-----96                     | 96  | Escherichia coli K-12                               |
| P44947 | 37  | ITFSKNAVRIASp-AVVNVYHRS---FSSASINDND-Q-LQVNLGSGvimsKdGylITkHLITQNAQI--VVAL      | 104 | Haemophilus influenza...                            |
| 066439 | 59  | PGGGPQRQHAD-----ALGS-----GFVISADg-YVVTNNHVIKge-dGNTv-aDE---ISIEFANGKLLDA        | 115 | Rhodobacter capsulatus                              |
| AS1TX7 | 37  | ENNVTKIQDTNI fpyTGVVAFKS-----ATGFVVGKIN--TILTNKHVSKn---KVGDR-iTAHPSDKNGG---IY51 | 102 | Staphylococcus aureus...                            |
| AS1TX6 | 38  | KIVTQKQDTNI fpyYGVVSKFD-----ATGVVIGKIN--TILTNKHVSkd---YKV---GD-RITAHPMWdKgNGI   | 99  | Staphylococcus aureus...                            |
| 20K5_A | 534 | FEevLFHPHRYNGKkeagipefydYVALIRKRNK-----LYGQGTIPICPtegttRaiRlPPTTCQQQE           | 685 | human                                               |
| Q2YMK6 | 175 | IG-----ADPRT---DLAVLKTNP-----KRFV-YVAFGD---Dn---KVRVSGWNVAV                     | 215 | Brucella abortus 2308                               |
| P26982 | 153 | KV---VG---KDPK-----DIALVKTQNP-----KINLTATKLADSD-----ALRVGYTVAT                  | 194 | Salmonella enterica s...                            |
| Q44596 | 140 | RD-----ET-----DLAVLKTIEAK-----QFPVLALGNSD-----EVEVGDLVAT                        | 177 | Brucella abortus                                    |
| P39668 | 145 | SR---VSADLVGSDQI-----DLAVLKRKSD-----KIKAVADFGNSD-----KVGSGEPVAT                 | 191 | Bacillus subtilis sub...                            |
| 3LGV_C | 97  | Rv---FEALVGSQSI-----DLAVLKTINAT-----GLPTTIPINARR-----VPHGQVVLAT                 | 143 | Escherichia coli K-12                               |
| P44947 | 105 | QNgIIFEALVGSQSI-----DLAVLKTIRAD-----NLSTIPQNSAR-----QAHVGQVVLAT                 | 153 | Haemophilus influenza...                            |
| 066439 | 116 | KIV---GTDPKT-----DIALLVKESK-----EPLPFVSGDSOT-----HREGQVNMAM                     | 157 | Rhodobacter capsulatus                              |
| AS1TX7 | 103 | KKI---IN---YPGK-----DVSIVQEERaiergpkGFNDNVPFKYAA---Ga---KAGERIKVIGY             | 157 | Staphylococcus aureus...                            |
| AS1TX6 | 100 | YKI---KSTSDYPPGD-----DISVWNIEEQaiergpkGFNERNVQAFYAA---Da---KVDDIKVIGY           | 157 | Staphylococcus aureus...                            |
| 20K5_A | 606 | ELIpaGDKTAlfveeeekLLTRKEVTVINMGKGSCEERdayAPGYdkvkdsVTPRFLCTGvsspydntICRGSG      | 685 | human                                               |
| Q2YMK6 | 216 | GN---PFGLG-----GTVTSIGVSALGRSD-IGAGP-----YDFITQIDA-----AVNRGNSG                 | 258 | Brucella abortus 2308                               |
| P26982 | 195 | GN---PFGLG-----ETVTSIGVSALGRSLINVEN-----YDFITQIDA-----AINRGNSG                  | 238 | Salmonella enterica s...                            |
| Q44596 | 178 | GN---PFGVG-----QTVTSIGVSASRTQVTSDF-----DEFITQIDA-----AINPGNSG                   | 221 | Brucella abortus                                    |
| P39668 | 192 | GhIglEFAG-----SVTQGVISGTERAIPVDSNg-dGQpd-----WNAEVLQIDA-----AINPGNSG            | 244 | Bacillus subtilis sub...                            |
| 3LGV_C | 144 | GN---PYNLG-----QTITGGLISATGRILHPT-----GR-----QNFITQIDA-----SINRGNSG             | 187 | Escherichia coli K-12                               |
| P44947 | 154 | GN---PYNLG-----QSVSGTISATGRNAGDSVg-----RQNFITQIDA-----SINRGNSG                  | 198 | Haemophilus influenza...                            |
| 066439 | 158 | GhIp---LG-----QGFSTSGVLVSA---RNRe-LSGTY-----DCCIQTIDA-----SINRGNSG              | 199 | Rhodobacter capsulatus                              |
| AS1TX7 | 158 | PHI---PYKNC-----YVLYESTGPNVSE-GSSIV-----YSAH---TES-----GNSG                     | 194 | Staphylococcus aureus...                            |
| AS1TX6 | 158 | Plp-pQISFK-----QFE-----STG-TIKR-----IDKRELNFDA-----YIEPGNSG                     | 194 | Staphylococcus aureus...                            |
| 20K5_A | 686 | GPLTHKRSrFIQGVISMGvDCKNQKqkvPAHA-RDFHInlfQVLPLH                                 | 737 | human                                               |
| Q2YMK6 | 259 | GPADLSG---EVIGINTAI---FSPSGGSVg---IAFA-IPSS---TAKQV                             | 298 | Brucella abortus 2308                               |
| P26982 | 239 | GALVNLG---ELIGINTAI---LAPOGGIlg---IGA-IPSN---MVKNL                              | 278 | Salmonella enterica subsp. enterica serovar Typh... |
| Q44596 | 222 | GALLIDRG---RLIGINTAI---YSRSGGSVg---IGA-IPSN---MVRVAV                            | 261 | Brucella abortus                                    |
| P39668 | 245 | GALLIDRG---KVTIGNSP---IAEAVG---IGLS-IPSN---LVIPDI                               | 283 | Bacillus subtilis subsp. subtilis str. 168          |
| 3LGV_C | 188 | GALVNSLG---ELMGINTLSF-DKSNIDGfne-gIGA-IPFG---LATK                               | 229 | Escherichia coli K-12                               |
| P44947 | 199 | GALINSAG---ELVGISTLSIgKTANIEAte---LNFA-IPID---IANDVL                            | 240 | Haemophilus influenzae Rd KW20                      |
| 066439 | 200 | GPLFSMDG---KVGIVNTAI---LSPNG---STG-IGFS---MASNV                                 | 235 | Rhodobacter capsulatus                              |
| AS1TX7 | 195 | SPVLNSNI---ELVGIFHAS---DKNIDDIIn---aGVYfTPEI---KKF                              | 234 | Staphylococcus aureus subsp. aureus JH9             |
| AS1TX6 | 195 | SPVLNSNI---ELVGIVYGGI-GKIGSE---YNGA-VYfTp---QIKDF                               | 233 | Staphylococcus aureus subsp. aureus JH9             |

https://www.ncbi.nlm.nih.gov/Taxonomy/Browser/wwwtax.cgi?id=71421

gh UV  
zw

As another approach, take the 10 ferritin light chain proteins from problem [6-1] and enter them into the MAFFT site at EBI (<http://www.ebi.ac.uk/Tools/msa/mafft/>) and setting the output format to ClustalW.

https://www.ebi.ac.uk/jdispatcher/msa/mafft?type=automatic&format=clustalw

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# MAFFT

Multiple Sequence Alignment (MSA)

Job Dispatcher Help & Privacy Your Jobs **Input form** [Feedback](#)

Welcome to the Job Dispatcher website! If you need assistance or have feedback, please [contact us](#).

MAFFT (Multiple Alignment using Fast Fourier Transform) is a high speed multiple sequence alignment program. We have recently changed the default parameter settings for MAFFT. Alignments should run much more quickly and larger DNA alignments can be carried out by default. Please click the 'More options' button to review the defaults and change them if required. Important note: This tool can align up to 500 sequences or a maximum file size of 1 MB.

**Input sequence ⓘ**

**Sequence type**  
☒ Automatic ☐ Protein ☐ DNA

**Paste your sequence here - or use the example sequence**

```

KTPDAMKAAAMALEKKLNQALLDLHALGSAHTDPLCDFLETHFLDEEVKLIKMGDHLTNLRLGGPEAGLGEYLFERLT
LKHD
>XP_010366918.1 ferritin light chain [Rhinopithecus roxellana]
MSSQIRQNVSTQVEAANSLNLYLQASYTYLSLGFYFDRODVALEGVSHFFRELAEEKREGYERLKMQNQRGRALFQ
DKKKPAEDWGWKTPDAMKAAAMALEKKLNQALLDLHALGSAHTDPLCDFLETHFLDEEVKLIKMGDHLTNLRLAGPEA
GLGEYLFERLT LKHD
>NP_001248136.1 ferritin light chain [Macaca mulatta]

```

[Choose File](#) [No file chosen](#) [Use the example](#) [Clear sequence](#) [More example inputs](#)

**Parameters**

**OUTPUT FORMAT ⓘ**  
 ClustalW

[More options ▼](#)

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The results:


<https://www.ebi.ac.uk/jdispatcher/msa/mafft/summary?jobId=mafft-l20250603-0620>

Alignment with colours

Hide

CLUSTAL format alignment by MAFFT FFT-NS-i (v7.525)

```

NP_000137.2      -----
XP_008958541.3  -----
NP_001126850.1  -----
XP_063672238.1  MFWQFGGPAGLSLASTVFGRRNRSGDSLPA$DRPPISSPLATSGTIFSAISCFWDLPA$FL
XP_010366918.1  -----
NP_001248136.1  -----
XP_011834729.1  -----
XP_023038337.1  -----
XP_032610358.1  -----
XP_050637471.1  M-----EQIRGLFLPASDRPPISSPLANS$GTIFSAISCFWDLPA$FL

NP_000137.2      -----MSSQIRQNYSTDEAAVNSLVNLYLQAS$TYLSLGFYFDRDDVALEGVSHF
XP_008958541.3  -----MSSQIRQNYSTDEAAVNSLVNLYLQAS$TYLSLGFYFDRDDVALEGVSHF
NP_001126850.1  -----MSSQIRQNYSTDEAAVNSLVNMYLQAS$TYLSLGFYFDRDDVALEGVSHF
XP_063672238.1  WLAP$CQPTMSSQIRQNYSTDEAAVNSLVNLYLQAS$TYLSLGFYFDRDDVALEGVSHF
XP_010366918.1  -----MSSQIRQNYSTDEAAVNSLVNLYLQAS$TYLSLGFYFDRDDVALEGVSHF
NP_001248136.1  -----MSSQIRQNYSTDEAAVNSLVNMYLQAS$TYLSLGFYFDRDDVALEGVSHF
XP_011834729.1  -----MSSQIRQNYSTDEAAVNSLVNMYLQAS$TYLSLGFYFDRDDVALEGVSHF
XP_023038337.1  -----MSSQIRQNYSTDEAAVNSLVNLYLQAS$TYLSLGFYFDRDDVALEGVSHF
XP_032610358.1  -----MSSQIRQNYSTDEAAVNSLVNLYLQAS$TYLSLGFYFDRDDVALEGVSHF
XP_050637471.1  WLAP$CRPTMSSQIRQNYSTDEAAVNSLVNMYLQAS$TYLSLGFYFDRDDVALEGVSHF
*****

NP_000137.2      FRELAEEKREGYERLLKMQNQRGGRALFQDIKKPAEDEW$KTPDAMKAAMALEKKLNQAL
XP_008958541.3  FRELAEEKREGYERLLKMQNQRGGRALFQDIKKPAEDEW$KTPDAMKAAMALEKKLNQAL
NP_001126850.1  FRELAEEKREGYERLLKMQNQRGGRALFQDIKKPAEDEW$KTPDAMKAAMALEKKLNQAL
XP_063672238.1  FRELAEEKREGYERLLKMQNQRGGRALFQDIKKPAEDEW$KTPDAMKAAMALEKKLNQAL
XP_010366918.1  FRELAEEKREGYERLLKMQNQRGGRALFQDVKKPAEDEW$KTPDAMKAAMALEKKLNQAL
NP_001248136.1  FRELAEEKREGYERLLKMQNQRGGRALFQDVKKPAEDEW$KTPDAMKAAMALEKKLNQAL
XP_011834729.1  FRELAEEKREGYERLLKMQNQRGGRALFQDVKKPAEDEW$KTPDAMKAAMALEKKLNQAL
XP_023038337.1  FRELAEEKREGYERLLKMQNQRGGRAFFQDVKKPAEDEW$KTPDAMKAAMALEKKLNQAL
XP_032610358.1  FRELAEEKRKGYERLLKMQNQRGGRALFQDIKKPAEDEW$KTPDAMKAAMALEKKLNQAL
XP_050637471.1  FRELAEEKREGYERLLKMQNQRGGRALFQDVKKPAEDEW$KTPDAMKAAMALEKKLIQAL
*****

NP_000137.2      LDHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGEYLFERLT
XP_008958541.3  LDHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGEYLFERLT
NP_001126850.1  LDHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGEYLFERLT
XP_063672238.1  LDHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGEYLFERLT
XP_010366918.1  LDHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNL$AGPEAGLGEYLFERLT
NP_001248136.1  LDHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNL$AGPEAGLGEYLFERLT
XP_011834729.1  LDHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNL$AGPEAGLGEYLFERLT
XP_023038337.1  LDHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNL$AGPEAGLGEYLFERLT
XP_032610358.1  LDHALGSACTDPHLCDFLETHFLDEEVKLIKMGDRLTNLHRLGGPEAGLGKYL$ERLT
XP_050637471.1  LDHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLNL$AGPEAGLGEYLFERLT
*****

NP_000137.2      LKHD
XP_008958541.3  LKHD
NP_001126850.1  LKHD
XP_063672238.1  LKHD
XP_010366918.1  LKHD
NP_001248136.1  LKHD
XP_011834729.1  LKHD
XP_023038337.1  LKHD
XP_032610358.1  LKHD
XP_050637471.1  LKHD
*****

```

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Search

For this particular set of proteins the EBI MUSCLE output is essentially the same but the order of the proteins is different.

https://www.ebi.ac.uk/dispatcher/msa/muscle?stype=protein

## Muscle

Multiple Sequence Alignment (MSA)

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Welcome to the Job Dispatcher website! If you need assistance or have feedback, please [contact us](#).

MUSCLE stands for Multiple Sequence Comparison by Log-Expectation. MUSCLE is claimed to achieve both better average accuracy and better speed than ClustalW2 or T-Coffee, depending on the chosen options. This tool can align up to 500 sequences or a maximum file size of 1 MB.

**Input sequence** Ⓢ **Paste your sequence here - or use the example sequence**

```
>NP_000137.2 ferritin light chain [Homo sapiens]
MSISQIRQNYSTGVDAANSLNLYLQADYTYLSLGFYFDRDOVALEGVSHFFRELAEKREGYERLLKMQNQRGRALFQ
DIKKPADEWIGKTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCCFLETHFLDEEVLIKQMGDHLNLHRLGGPEA
GLGEVYFERLTUXHD
>XP_008585412 ferritin light chain [Pan paniscus]
MSISQIRQNYSTGVDAANSLNLYLQADYTYLSLGFYFDRDOVALEGVSHFFRELAEKREGYERLLKMQNQRGRALFQ
DIKKPADEWIGKTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCCFLETHFLDEEVLIKQMGDHLNLHRLGGPEA
...

```

[Choose file](#) [No file chosen](#) [Use the example](#) [Clear sequence](#) [More example inputs](#)

**Parameters** **OUTPUT FORMAT** Ⓢ

ClustalW

[More options](#)

**Submit** **Title**

Muscle's job

[Submit](#)

If you use this service, please consider citing the following publication: [The EMBL-EBI Job Dispatcher sequence analysis tools framework in 2024](#). More information about this bioinformatics application can be found in its [bio.tools](#) record.

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EMBL-EBI is the home for big data in biology.

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The results:


<https://www.ebi.ac.uk/jdispatcher/msa/muscle/summary?jobId=muscle-l20250603-06>

Alignment with colours

Hide

CLUSTAL multiple sequence alignment by MUSCLE (3.8)

```

XP_032610358.1 -----
XP_063672238.1 MFWQFGGPAGLSLASTVFGRRNSGDSLPASDRPPISSPLATSGTIFSAISCFWDLAPFL
NP_001126850.1 -----
NP_000137.2 -----
XP_008958541.3 -----
XP_050637471.1 -----MEQIRGLFLPASDRPPISSPLANSGTIFSAISCFWDLAPFL
XP_011834729.1 -----
NP_001248136.1 -----
XP_023038337.1 -----
XP_010366918.1 -----

XP_032610358.1 -----MSSQIRQNYSTOVEAAVNSLVNLYLQASYYTSLSGFYFDRDDVALEGVSHF
XP_063672238.1 WLAPSCQPTMSSQIRQNYSTOVEAAVNSLVNLYLQASYYTSLSGFYFDRDDVALEGVSHF
NP_001126850.1 -----MSSQIRQNYSTOVEAAVNSLVNMYLQASYYTSLSGFYFDRDDVALEGVSHF
NP_000137.2 -----MSSQIRQNYSTOVEAAVNSLVNLYLQASYYTSLSGFYFDRDDVALEGVSHF
XP_008958541.3 -----MSSQIRQNYSTOVEAAVNSLVNLYLQASYYTSLSGFYFDRDDVALEGVSHF
XP_050637471.1 WLAPSCQPTMSSQIRQNYSTOVEAAVNSLVNMYLQASYYTSLSGFYFDRDDVALEGVSHF
XP_011834729.1 -----MSSQIRQNYSTOVEAAVNSLVNMYLQASYYTSLSGFYFDRDDVALEGVSHF
NP_001248136.1 -----MSSQIRQNYSTOVEAAVNSLVNMYLQASYYTSLSGFYFDRDDVALEGVSHF
XP_023038337.1 -----MSSQIRQNYSTOVEAAVNSLVNLYLQASYYTSLSGFYFDRDDVALEGVSHF
XP_010366918.1 -----MSSQIRQNYSTOVEAAVNSLVNLYLQASYYTSLSGFYFDRDDVALEGVSHF

*****

XP_032610358.1 FRELAEEKRKGYERLLKMQNRGGRAFQDIKKPAEDEWGKTPDAMKAAMALEKKLNQAL
XP_063672238.1 FRELAEEKREGYERLLKMQNRGGRAFQDIKKPAEDEWGKTPDAMKAAMALEKKLNQAL
NP_001126850.1 FRELAEEKREGYERLLKMQNRGGRAFQDIKKPAEDEWGKTPDAMKAAMALEKKLNQAL
NP_000137.2 FRELAEEKREGYERLLKMQNRGGRAFQDIKKPAEDEWGKTPDAMKAAMALEKKLNQAL
XP_008958541.3 FRELAEEKREGYERLLKMQNRGGRAFQDIKKPAEDEWGKTPDAMKAAMALEKKLNQAL
XP_050637471.1 FRELAEEKREGYERLLKMQNRGGRAFQDVKKPAEDEWGKTPDAMKAAMALEKKLIQAL
XP_011834729.1 FRELAEEKREGYERLLKMQNRGGRAFQDVKKPAEDEWGKTPDAMKAAMALEKKLNQAL
NP_001248136.1 FRELAEEKREGYERLLKMQNRGGRAFQDVKKPAEDEWGKTPDAMKAAMALEKKLNQAL
XP_023038337.1 FRELAEEKREGYERLLKMQNRGGRAFQDVKKPAEDEWGKTPDAMKAAMALEKKLNQAL
XP_010366918.1 FRELAEEKREGYERLLKMQNRGGRAFQDVKKPAEDEWGKTPDAMKAAMALEKKLNQAL

*****

XP_032610358.1 LDHALGSACTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGKYLFERLT
XP_063672238.1 LDHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGKYLFERLT
NP_001126850.1 LDHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGKYLFERLT
NP_000137.2 LDHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGKYLFERLT
XP_008958541.3 LDHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGKYLFERLT
XP_050637471.1 LDHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGKYLFERLT
XP_011834729.1 LDHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGKYLFERLT
NP_001248136.1 LDHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGKYLFERLT
XP_023038337.1 LDHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGKYLFERLT
XP_010366918.1 LDHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEAGLGKYLFERLT

*****

XP_032610358.1 LKHD
XP_063672238.1 LKHD
NP_001126850.1 LKHD
NP_000137.2 LKHD
XP_008958541.3 LKHD
XP_050637471.1 LKHD
XP_011834729.1 LKHD
NP_001248136.1 LKHD
XP_023038337.1 LKHD
XP_010366918.1 LKHD

****

```

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Search

To see differences between MSA programs it is helpful to choose more divergent sequences.  
Try adding these:

>gi|91081285|ref|XP\_967895.1| PREDICTED: soma ferritin [Tribolium castaneum]

MAQSQVRQNFHKDCEDAINKQINVELNAFYTYLSMAYHFQRDDVALPGLYKYFKACSDDEERDHAHKLMEYLNKRGGRLAL  
TDIPAPEKQDWGTAQEAMCAALDLEKRVNESLLVLHSTASGHMDVNLCDLFLETHYLQEQQVDAIKEIADHVTNLKRVGEG  
GVYMFDRTLADE

>gi|24641673|ref|NP\_572854.1| ferritin 3 heavy chain homologue [Drosophila melanogaster]

MAWCFRDIRRHMCMLVRQNFASCEKKLNDQINMELKASHQYLAMAYHFDRSDISSPGMHRFFLKASVEEREHA EKIMTY  
MKNRGGGLIILSSVPQPLPCFASTLDALKHAMKMELEVNHLLDLHALAGKEADPNLCDFIEANFLQEQQVDGQKILADYIS  
QLEKAQNQVGEFLFDKYMGS GMHPAK

>gi|453232411|ref|NP\_504944.2| FTN-1 [Caenorhabditis elegans]

MSLARQNYHDEVEAAVNKQINVELYASYVYLSMSAHFDRDDIALRNIAKFFKEQSDEERGHATELMRIQAVRGGRVAMQN  
IQKPEKDEWGTVLEAFEALALERANNASLLKLHGIAEQRND AHLTNYIQEKYLEEQVHSINEFARYIANIKRAGPGLGE  
YLFDKKEEFS

>gi|524888390|ref|XP\_005100766.1| PREDICTED: ferritin light chain, oocyte isoform-like  
isoform X1 [Aplysia californica]

MTNRGFVLFCVLKMSLHTGVLLILMVTISSIQPTDYSRDANVEISSRCQKRVKRCGQFKPPKPEMGEMSCSFSARTCGFA  
CDLICPTPEGKTALVLNQKRHGRD MYLCTAGNWKRRTPQTFCQAQWPVKEVRQNLHHVENLNGLVNKLNTSYFYLGMA  
SFYERADVALPGFSKLM TDLWSADQSHARDLMSYVNKRGGYITLYDIPRTPSHEVLLKLSSRIGQAGVEMALQAAREVN  
EQVLELHK NATLPGDSNDPHLKHALEDGLSSKVELIKKLADVDRRLHAFPEEDYAVGEYVLDQEQLG

Rename them:

>beetle XP\_967895.1

MAQSQVRQNFHKDCEDAINKQINVELNAFYTYLSMAYHFQRDDVALPGLYKYFKACSDDEERDHAHKLMEYLNKRGGRLAL  
TDIPAPEKQDWGTAQEAMCAALDLEKRVNESLLVLHSTASGHMDVNLCDLFLETHYLQEQQVDAIKEIADHVTNLKRVGEG  
GVYMFDRTLADE

>fly NP\_572854.1

MAWCFRDIRRHMCMLVRQNFASCEKKLNDQINMELKASHQYLAMAYHFDRSDISSPGMHRFFLKASVEEREHA EKIMTY  
MKNRGGGLIILSSVPQPLPCFASTLDALKHAMKMELEVNHLLDLHALAGKEADPNLCDFIEANFLQEQQVDGQKILADYIS  
QLEKAQNQVGEFLFDKYMGS GMHPAK

>nematode NP\_504944.2

MSLARQNYHDEVEAAVNKQINVELYASYVYLSMSAHFDRDDIALRNIAKFFKEQSDEERGHATELMRIQAVRGGRVAMQN  
IQKPEKDEWGTVLEAFEALALERANNASLLKLHGIAEQRND AHLTNYIQEKYLEEQVHSINEFARYIANIKRAGPGLGE  
YLFDKKEEFS

>Aplysia XP\_005100766.1

MTNRGFVLFCVLKMSLHTGVLLILMVTISSIQPTDYSRDANVEISSRCQKRVKRCGQFKPPKPEMGEMSCSFSARTCGFA  
CDLICPTPEGKTALVLNQKRHGRDMYLCTAGNWKRRTPQTFCKAQWPVKEVRQNLHHVENLNLVKNLLNTSYFYLGMA  
SFYERADVALPGFSKLM TDLWSADQSHARDLMSYVNKRGGYITLYDIPRTPSHEVLLLKLSSRIGQAGVEMALQAAREVN  
EQVLELHK NATLPGDSNDPHLKHAE DGLSSKVELIKKLADVD RRLHAFPEEDYAVGEYVLDQEQLG

Add them, creating a set of 9 ferritins:

>human NP\_000137.2

MSSQIRQNYSTDVEAAVNSLVNLYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQNQRGGRALFQ  
DIKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLGGPEA  
GLGEYLFERLTLKHD

>bonobo XP\_008965255.1

MTQQSLPATRTPDTGLAGHLQIPSNQPQGNPPKPPFHNTLALQARDFPLLWGGGLRLLCAPDWSGTAFGPASCRRRLAA  
SPPRAPCLRGAHYKRSRPSHVPSQFGGPAGLSLASTVFGRNRSGLSPASDCPPISSPLATSGTIFSAISCFWDLPAF  
LWLAPSCQPTMSSQIRQNYSTDVEAAVNSLVNLYLQASYTYLSLGFYFDRDDVALEGVSHFFRELAEEKREGYERLLKMQ  
NQRGGRALFQDIKKPAEDEWGKTPDAMKAAMALEKKLNQALLDLHALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLT  
NLHRLGGPEAGLGEYLFERLTLKHD

>mouse NP\_034370.2

MTSQIRQNYSTEVEAAVNRLVNLHLRASITYLSLGFYFDRDDVALEGVGHFFRELAEEKREGAERLLEFQNDRGGRALFQ  
DVQKPSQDEWGKTQEAMEAALAMEKNLNQALLDLHALGSARADPHLCDFLESHYLDKEVKLIKMGNHLTNLRRVAGPQP  
AQTGAPQGSLGEYLFERLTLKHD

>horse XP\_005602735.1

MSSHIRQNYSTEVEAAINRLVNLHLRASITYLSLGFYFNRRDDVALEGVCHFFCELAEEKRECAKCLLKMQNQHGDHALFQ  
DLQKPSQDEWGTTLDAMKAAVVLEKSLNQALLDLHALGSAHADPHLCDFLESHFLDEEVKLIKMGDNLTNIQRLVGPQA  
GLGECLFERLTLKHD

>dog XP\_005621156.1

MSSRIRQNYSTEVEAVVNRLVNMHLRASITYLSLGFYFDRDDVALEGVGHFFRELEKCEGAERFLKMQNQRGGCALFQDV  
QKPSQDEWGKTL DAMEAALLLEKNLNQALLDLHALGSARADPHLCDFLENHFLDEVKLIKMGDHLTNLRRLATPQAGLG  
EYLFERLTLKHD

>beetle XP\_967895.1

MAQSQVRQNFHKDCEDAINQINVELNAFYTYLSMAYHFQRDDVALPGLYKYFKACSD EERDHAHKLMEYLNKRGGRLAL  
TDIPAPEKQDWGTAQEAMCAALDLEKRVNESLLVLHSTASGHMDVNLCDFLETHYLQEQQVDAIKEIADHVTNLKRVGEGL  
GVYMFDRTLADE



>fly NP\_572854.1

MAWCFDIRRHMCMLVRQNFAKSCEKKLNDQINMELKASHQYLAMAYHFDSDISSPGMHRFFLKASVEEREHA EKIMTY  
MNRKGGLIILSSVPQPLPCFASTLDALKHAMKMELEVNKHLLDLHALAGKEADPNLCDFIEANFLQEVDGQKILADYIS  
QLEKAQNQVGEFLFDKYMGSGMHPAK

>nematode NP\_504944.2

MSLARQNYHDEVEAAVNKQINVELYASYVYLSMSAHFDRDDIALRNIAKFFKEQSDEERGHATELMRIQAVRGGRVAMQN  
IQKPEKDEWGTVLEAFEAAALALERANNASLLKLHGIAEQRND AHLTNYIQEKYLEEQVHSINEFARYIANIKRAGPGLGE  
YLFDKKEEFS

>Aplysia XP\_005100766.1

MTNRGFVLFCVLKMSLHTGVLLILMVTISSIQPTDYSRDANVEISSRCQKRVKRCGQFKPPKPEMGEMSCSFSARTCGFA  
CDLICPTPEGKTALVNLQKRHRGRD MYLCTAGNWKRRPTQTFCKAQWPVKEVRQNLHHVENLNLV NKLNTSYFYLGMA  
SFYERADVALPGFSKLM TDLWSADQSHARDLMSYVNKRGGYITLYDIPRTPSHEVLLKLSSRIGQAGVEMALQAAREVN  
EQVLELHK NATLPGDSNDPHLKHAE DGLSSKVELIKKLADVDRRLHAFPEEDYAVGEYVLDQEQLG

View the EBI MAFFT alignment:

The screenshot shows the EBI MAFFT web interface. The browser address bar displays the URL: <https://www.ebi.ac.uk/jdispatcher/msa/mafft?type=automatic&format=clustalw>. The page has a blue header with the MAFFT logo and navigation links. A yellow banner at the top contains a welcome message. The main content area is divided into sections: 'Input sequence', 'Parameters', and 'Submit'. The 'Input sequence' section has a 'Sequence type' dropdown set to 'Automatic' and a text area containing the three sequences from the previous blocks. Below the text area are buttons for 'Choose file', 'Use the example', 'Clear sequence', and 'More example inputs'. The 'Parameters' section has an 'OUTPUT FORMAT' dropdown set to 'ClustalW' and a 'More options' link. The 'Submit' section has a 'Title' text field with the value 'MAFFT's job' and a 'Submit' button. At the bottom, there is a footer with a citation notice and a system tray showing the date and time as 1:33 PM on 3/6/2025.

The results:

Alignment with colours

Hide

CLUSTAL format alignment by MAFFT FFT-NS-1 (v7.525)

```

human      -----
bonobo     MTQQS-----LPATRTPTDGLAGHLQIPSNQPGPNPPKPPFHNTLALQAHDFLLWGG
mouse      -----
dog         -----
horse      -----
beetle     M-----
nematode   M-----
fly        MA-----
Aplysia    MTNRGFVLCVLKMSLHTGVLLILMVITISSTQPT-----

human      -----
bonobo     GLRLLCAPDWSGTAFGPAS--CRRRLAASPPRAPCLGPAHYKRSRPSHPVSQFGGPAGL
mouse      -----
dog         -----
horse      -----
beetle     -----
nematode   -----
fly        -----
Aplysia    -----W-----C-----GQFKPPKPEM-----GEMSC

human      -----
bonobo     SLASTVFGRRHSGGSLPASD--CPPISPLATSGTIFSAISCFWDL-----P
mouse      -----
dog         -----
horse      -----
beetle     -----
nematode   -----
fly        -----CFHDT-----RRH--
Aplysia    SFSARTCGF-----ACOLICP--TPEGKTALVLNQKRHGRDHYLCTAGNMKRRRTEP

human      -----
bonobo     -----MSSQIRQNYSTDEAAVNSLVNLYLQASYTYLSLGFYFDRDDVALE
mouse      -----MTSQIRQNYSTEVEAAVNRVLNHLRASYTYLSLGGFFDRDDVALE
dog         -----MSSIRQNYSTEVEAAVNRVLNHLRASYTYLSLGFYFDRDDVALE
horse      -----MSSHIRQNYSTEVEAAINRVLNLYRASYTYLSLGFYFNDDVALE
beetle     -----AQSQVRQNFHKDCEDATNKQINVELNAFYTYLSMAYHFQDDVALP
nematode   -----SLARQNYHDEVEAAVNRQINVELYASYTYLSMSAHFDRDDIALR
fly        -----MCMLVRQNFASCEKKLNDQINMELKASHQYLAMAYHFDRSDISSP
Aplysia    QTF-----CKAQWPVKEVRQNLH-HVEN-LNGLVNLKLLNTSYFYLGMSFYERADVALP
                ***      *  : *  : *  : *  : *  : *  : *  : *  :

human      GVSHFFRELAEKREGYERLLKMQNQGGRALFQDIKK-PAEDWNGKTPDA-----MKA
bonobo     GVSHFFRELAEKREGYERLLKMQNQGGRALFQDIKK-PAEDWNGKTPDA-----MKA
mouse      GVCHFFRELAEKREGAERLLEFQNDRGGRALFQDVQK-PSQDEWNGKTEA-----MEA
dog         GVCHFFREL--EKCEGAERFLKMQNQGGCALFQDVQK-PSQDEWNGKTLDA-----MEA
horse      GVCHFFCELAEKRECAKCLLKMNQHQGDHALFQDLQK-PSQDEWNGTTLDA-----MKA
beetle     GLYKYFKACSDDEERDHAHLKNEYLNRGGRLALTDPAP-PEKQWNGTAQEA-----MCA
nematode   NIAKFFKEQSDDEERGHAELNRIQAVRGGVAMQNTQK-PEKDEWNGTLEA-----FEA
fly        GMRFFFLKASVEEREHAEKINTYMNKRGGLITLSSVPQ-P-LPCFASTLDA-----LKH
Aplysia    GFSKLMTDLWSAQSHARDLNSYVNRGGYITLYDTPRTPSHEVLLKLLSSRIQAGVEM
                .. : :  .  : :  : *  : :  : *  : :  :

human      AMALEKKLNQALLDLH---ALGSARTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLG
bonobo     AMALEKKLNQALLDLH---ALGSAHTDPHLCDFLETHFLDEEVKLIKMGDHLTNLHRLG
mouse      ALAMEKLNQALLDLH---ALGSARADPHLCDFLESHYLDKEVKLIKMGHNLNLRRVA
dog         ALLLEKLNQALLDLH---ALGSARADPHLCDFLENHFLD-EVKLIKMGDHLTNLRLA
horse      AVVLEKSLNQALLDLH---ALGSAHADPHLCDFLESHFLDEEVKLIKMGDNLNTIQRVY
beetle     ALDLEKRVNESLLVLH---STASGHMDVNLCDFLETHYEQVDKIEADHVTNLKRVG
nematode   ALALERAMNASLLKLH---GIAEQRND AHLTNYIQEKYLEEQVHSTNEFARYIANIKRAG
fly        AMKMELEVNHLLDLH---ALAGKEADPNLCDFTEANFLQEQVGGQKILADYISQLEKAQ
Aplysia    ALQAAREVNEQVLELHNATLPGDSNDPHLXHALEDGLSSKVELIKKLADVDRLHAPP
                * :      *  : *  *      *  : *  :  : *  : *  :  :  :

human      G-----PEAGLGEYLFERLTLYK----HD
bonobo     G-----PEAGLGEYLFERLTLYK----HD
mouse      GPQPAQTGAPQGSGLGEYLFERLTLYK----HD
dog         T-----PQAGLGEYLFERLTLYK----HD
horse      G-----PQAGLGEYLFERLTLYK----HD
beetle     -----EGLGVYMFORTLAD-----E
nematode   -----PGLGEYLFQKEEFS-----D
fly        -----NQVGEFLFKYMGSGMHPAK
Aplysia    E-----EDYAVGEYVLQEQLE-----
                : *  : : : :

```

Then compare MUSCLE.

https://www.ebi.ac.uk/jdispatcher/msa/muscle?style=protein

### Muscle

Multiple Sequence Alignment (MSA)

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MUSCLE stands for Multiple Sequence Comparison by Log-Expectation. MUSCLE is claimed to achieve both better average accuracy and better speed than ClustalW2 or T-Coffee, depending on the chosen options. This tool can align up to 500 sequences or a maximum file size of 1 MB.

**Input sequence**

Paste your sequence here - or use the example sequence

```
IQIPKDEWVYLEAPFAALALHANNASLLKHGIAEQVINDAPLNTYQKYLEEQVPSINERARYIANKHAGPGLLE  
YLFKEEFD  
>Aplysia XP_005100766.1  
MTNKGFLVFLKMSLHTGVLLLVYTSISQPTDYSRANVEISSRQKQKRVKRCQKPKPKPEHNGMSQSPARTGPA  
GLLEPTREGLTALVKNQKSHGDMYCTAGAWKSRBTQTFQCAQWPKVSRQNLNHHVNLKGLWELLNTSYVFLGMA  
SPYERADIALPGFSKMLTDLWSADQSHARDLSYVKNRGQYTLVDPRTPSEVLLKLSRRIGQAGVEMALQAAREVN  
EQVLELHNATLPGDSNDPLKHAEGLGLSSKVELIKLADVDRLHAFPEEDYAVGEVLDQEQLG
```

[Choose File](#) [No file chosen](#) [Use the example](#) [Clear sequence](#) [More example inputs](#)

**Parameters**

OUTPUT FORMAT

ClustalW

[More options](#)

**Submit**

Title

Muscle's job

[Submit](#)

If you use this service, please consider citing the following publication: The EMBL-EBI Job Dispatcher sequence analysis tools framework in 2024. More information about this bioinformatics application can be found in its [bio.tools](#) record.

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3/6/2025

The results:

Hide

CLUSTAL multiple sequence alignment by MUSCLE (3.8)

[illegible]

Q Se

### 3) Use EDirect to access sets of homologous proteins from HomoloGene. These can be viewed in various formats.

- i. Retrieve the sets of protein sequences, in the FASTA format, containing the protein HBB. How many HomoloGene entries are there in this set?

#### (3) SOLUTIONS/COMMENTS:

```
$ esearch -db homologene -query "HBB" | efetch -db homologene -format fasta
```

1: HomoloGene:128037. Gene conserved in Boreoeutheria

```
>gi|4504351|ref|NP_000510.1| hemoglobin subunit delta
```

```
MVHLTPEEKTAVNALWGKVNVDVAVGGEALGRLLVVYPWTQRFFESFGDLSSPDVAVMGNPKVKAHGKKVLG
AFSDGLAHLNLTGTFSQLSEHCDKLHVDPENFRLLGNVLCVLAARNFGKEFTPQMQAAYQKVVAGVAN
ALAHKYH
```

```
>gi|332835679|ref|XP_001162045.2| PREDICTED: hemoglobin subunit delta isoform 1 [Pan
troglodytes]
```

```
MVHLTPEEKTAVNALWGKVNVDVAVGGEALGRLLVVYPWTQRFFESFGDLSSPDVAVMGNPKVKAHGKKVLG
AFSDGLAHLNLTGTFSQLSEHCDKLHVDPENFRLLGNVLCVLAARNFGKEFTPQVQAAYQKVVAGVAN
ALAHKYH
```

...

Next retrieve the alignment scores (rather than the sequences) for HomoloGene entries containing the protein HBB.

```
$ esearch -db homologene -query "HBB" | efetch -db homologene -format alignmentscores
```

1: HomoloGene:128037. Gene conserved in Boreoeutheria

Pairwise Alignment Scores

Gene Identity (%)

Species Symbol Protein DNA

H.sapiens HBD

vs. P.troglodytes HBD 99.3 99.3 Blast

vs. B.taurus HBB 83.3 86.1 Blast

vs. B.taurus HBG 77.8 82.6 Blast

vs. B.taurus LOC788610 77.8 82.6 Blast

P.troglodytes HBD

vs. H.sapiens HBD 99.3 99.3 Blast

vs. B.taurus HBB 83.3 85.9 Blast

vs. B.taurus HBG 77.8 82.4 Blast

vs. B.taurus LOC788610 77.8 82.4 Blast

...

#### 4) We described how ClustalW applies a correction factor to downweight the influence of closely related proteins. Test the performance of ClustalW.

- i. Take the globins in web documents 6.1 and/or 6.2 and align.
- ii. Then repeat the alignment with the additional input one divergent sequence repeated varying number of times. For example, in the closely related group of beta globins, add five copies of the chicken sequence to see its influence on the alignment.

##### (4) SOLUTIONS/COMMENTS:

Here is a set of distantly related globins; it is helpful to include some distantly related proteins.

>beta\_globin 2hhbB NP\_000509.1 [Homo sapiens]

```
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMVGNPKVKAHGKKVLG
AFSDGLAHLNLTGKTFATLSSEHCDKLHVDPENFRLLGNVLCVLAHHFGKEFTPPVQAAYQKVVAGVAN
ALAHKYH
```

>myoglobin 2MM1 NP\_005359.1 [Homo sapiens]

```
MGLSDGEWQLVLNVWGKVEADIPGHGQEVLRIRLFGKHPETLEKFDKFKHLKSEDEMKASEDLKKHGATVL
TALGGILKKKGHHEAEIKPLAQSHATKHKIPVKYLEFISECIIQVLQSKHPGDFGADAQGAMNKALELFR
KDMASNYKELGFQG
```

>neuroglobin 1OJ6A NP\_067080.1 [Homo sapiens]

```
MERPEPELIRQSWRAVSRSPLEHGTVLFARLFALPDLLPLFQYNCRQFSSPEDCLSSPEFLDHIRKVML
VIDAAVTNVEDLSSLEEYLASLGRKHRAVGKLSSTVSTVGESLLYMLEKCLGPAFTPATRAAWSQLYGAV
VQAMSRGWDGE
```

>soybean\_globin 1FSL leghemoglobin P02238 LGBA\_SOYBN [Glycine max]

```
MVAFTEKQDALVSSSEAFKANIPQYSVVFYTSILEKAPAAKDLFSFLANGVDPTNPKLTGHAELKLFALV
RDSAGQLKASGTVVADAALGSHVHAQKAVTDPQFVVVKEALLKTIKAAVGDKWSDELSRAWEVAYDELA
IKKA
```

>rice\_globin 1D8U rice Non-Symbiotic Plant Hemoglobin NP\_001049476.1 [Oryza sativa (japonica cultivar-group)]

```
MALVEDNNAVAVSFSEEQALVLKSWAILKKDSANIALRFFLKIFEVAPSASQMFSFLRNSDVPLEKNPK
LKTHAMSVFVMTCEAAAQLRKAGKVTVRDITLKRGLGATHLYGVGDAHFEVVKFALLDTIKEEVPA
PAMKSAWSEAYDHLVAAIKQEMKPAE
```

https://www.ebi.ac.uk/jdispatcher/msa/mafft?type=automatic&format=dustalw

# MAFFT

Multiple Sequence Alignment (MSA)

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MAFFT (Multiple Alignment using Fast Fourier Transform) is a high speed multiple sequence alignment program. We have recently changed the default parameter settings for MAFFT. Alignments should run much more quickly and larger DNA alignments can be carried out by default. Please click the 'More options' button to review the defaults and change them if required. Important note: This tool can align up to 500 sequences or a maximum file size of 1 MB.

**Input sequence** Sequence type

☒ Automatic ☐ Protein ☐ DNA

**Paste your sequence here - or use the example sequence**

```
>beta_globin 2hhbB NP_000509.1 [Homo sapiens]
MVHLTPEEKSAVTALWGKVNVDVEGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGNPKVKAHGKKV
AFSDGLAHLNLIKGTFTATLSELHCDKLHVDPENFRLLGNLVLCVLAHHFGKEFTPPVQAAAYQKV
ALAHKYH
>myoglobin 2MM1 NP_005359.1 [Homo sapiens]
MGLSDGEWQLVLNVWGKVEADIPGHGQEVLRIRFKGHPETLEKFDKFKHLKSEDEMKASEDLKKHG
TALGGILKKKGHHEAIEKPLAQSHATKHIPVKYLEFISECTIQVLQSKHPGDFGADAQGAMNKALELFR
```

[Choose File](#) [No file chosen](#) [Use the example](#) [Clear sequence](#) [More example inputs](#)

**Parameters** OUTPUT FORMAT

ClustalW [More options](#)

## The results:

https://www.ebi.ac.uk/jdispatcher/msa/mafft/summary?jobId=mafft-l20250603-093543-0920-46967542-p1m&js=pass

**Alignment with colours**

[Hide](#)

CLUSTAL format alignment by MAFFT FFT-NS-i (v7.525)

```
beta_globin      M-----VHLTPEEKSAVTALWGKVNVD--EVGGEALGRLLVVYPWTQRFFES-FG
neuroglobin      M-----ERPEPELIRQSWRAVSRSPLHGTVLFAFLFALEPDLLPLFYQWCR
soybean_globin   M-----VAFTEKQDALVSSSEAFKANIQQYSVVFTSILEKAPAAKDLFSF-LA
rice_globin      MALVEDNNNAVAVSFSEEQALVLKSWAILKKDSANIALRFFLKIFEVAPASQMFSE-LR
myoglobin        M-----GLSDGEWQLVLNVWGKVEADIPGHGQEVLRIRFKGHPETLEKFDK-FK
*               :   :   :   :   :   :   :   :   :   :   :   :   :
beta_globin      DLSTPDAMGNPKVKAHGKKVLAFAFSDGLAHLNLIK-----GTFATLSELHCDKLHVD
neuroglobin      QFSSPEDCLSSPEFLDHIRKVMVLIDAAVTNVEDLSLE---EYLASLGRKH-RAVGKL
soybean_globin   NGVDP---TNPKLTGHAELFALVRDSAGQLKASGTV---VADAALGSVH-AQKAVTD
rice_globin      NSDVP--LEKNPKLTHAMSVFVMTCEAAQLRKAGKVTVRDITLKRGLATH-LKYGVGD
myoglobin        HLKSEDEMKASEDLKKHGATVLTALG---GILKKKGHHE---AEIKPLAQSHATKHIPV
.               .   .   *   .   :   :   :   :   :   :   :   :   :
beta_globin      ENFRLLGNLVLCVLAHHF-GKEFTPPVQAAAYQKVAVGVANALAHKYH-----
neuroglobin      SSFSTVGESLLYMLEKCL-GPAFTPATRAAWSQLYGAVVQAMSRGWD----GE
soybean_globin   PQFVVVKEALLKTIKAAV-GDKWSELSRAWEVAYDELAALAIKKA-----
rice_globin      AHFEVVKFALLDTIKEEVPADMMSPAKSAWSEAYDHLVAAIKQEMKP---AE
myoglobin        KYLEFISECTIQVLQSKHPGD-FGADAQGAMNKALELFRKDMASNYKELGFQG
:   :   :   :   :   :   :   :   :   :   :   :   :   :   :
```

Note the insertion in rice globin (corresponding to a gap in the other four proteins) about halfway through the alignment. Add extra copies of neuroglobin (same sequence, unique names, in this case neuroglobin2, neuroglobin3, neuroglobin4):

>beta\_globin 2hhbB NP\_000509.1 [Homo sapiens]

```
MVHLTPEEKSAVTALWGKVNVDVEGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGNPKVKAHGKKV
AFSDGLAHLNLIKGTFTATLSELHCDKLHVDPENFRLLGNLVLCVLAHHFGKEFTPPVQAAAYQKV
ALAHKYH
```

>myoglobin 2MM1 NP\_005359.1 [Homo sapiens]

MGLSDGEWQLVLNVWGKVEADIPGHGQEVLRIRLFKGPETLEKFDKFKHLKSEDEMKASEDLKKHGATVL  
TALGGILKKKGHHEAEIKPLAQSHATKHKIPVKYLEFISECIIQVLQSKHPGDFGADAQGAMNKALELFR  
KDMASNYKELGFQG

>neuroglobin 1OJ6A NP\_067080.1 [Homo sapiens]

MERPEPELIRQSWRAVSRSPLEHGTVLFARLFALPDLLPLFQYNCRQFSSPEDCLSSPEFLDHIRKVML  
VIDAAVTNVEDLSSLEEYLASLGRKHRAVGVKLSSFSTVGESLLYMLEKCLGPAFTPATRAAWSQLYGAV  
VQAMSRGWDGE

>neuroglobin2

MERPEPELIRQSWRAVSRSPLEHGTVLFARLFALPDLLPLFQYNCRQFSSPEDCLSSPEFLDHIRKVML  
VIDAAVTNVEDLSSLEEYLASLGRKHRAVGVKLSSFSTVGESLLYMLEKCLGPAFTPATRAAWSQLYGAV  
VQAMSRGWDGE

>neuroglobin3

MERPEPELIRQSWRAVSRSPLEHGTVLFARLFALPDLLPLFQYNCRQFSSPEDCLSSPEFLDHIRKVML  
VIDAAVTNVEDLSSLEEYLASLGRKHRAVGVKLSSFSTVGESLLYMLEKCLGPAFTPATRAAWSQLYGAV  
VQAMSRGWDGE

>neuroglobin4

MERPEPELIRQSWRAVSRSPLEHGTVLFARLFALPDLLPLFQYNCRQFSSPEDCLSSPEFLDHIRKVML  
VIDAAVTNVEDLSSLEEYLASLGRKHRAVGVKLSSFSTVGESLLYMLEKCLGPAFTPATRAAWSQLYGAV  
VQAMSRGWDGE

>soybean\_globin 1FSL leghemoglobin P02238 LGBA\_SOYBN [Glycine max]

MVAFTEKQDALVSSSFEAFKANIPQYSVVFYTSILEKAPAAKDLFSFLANGVDPTNPKLTGHA EKLFALV  
RDSAGQLKASGTVVADAALGVSVAQKAVTDPQFVVVKEALLKTIKAAVGDKWSDELSRAWEVAYDELA  
IKKA

>rice\_globin 1D8U rice Non-Symbiotic Plant Hemoglobin NP\_001049476.1 [Oryza sativa (japonica cultivar-group)]

MALVEDNNAVAVSFSEEQALVLKSWAILKKDSANIALRFFLKIFEVAPSASQMFSFLRNSDVPLEKNPK  
LKTHAMSVFVMTCEAAAQLRKAGKVTVRDTTLKRLGATHLKYGVGDAHFEVVKFALLDTIKEEVPADMWS  
PAMKSAWSEAYDHLVAAIKQEMKPAE

Then, re-analyze in Clustal2W:

**MAFFT**  
Multiple Sequence Alignment (MSA)

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MAFFT (Multiple Alignment using Fast Fourier Transform) is a high speed multiple sequence alignment program. We have recently changed the default parameter settings for MAFFT. Alignments should run much more quickly and larger DNA alignments can be carried out by default. Please click the 'More options' button to review the defaults and change them if required. Important note: This tool can align up to 500 sequences or a maximum file size of 1 MB.

**Input sequence**

**Sequence type**  
☒ Automatic ☐ Protein ☐ DNA

**Paste your sequence here - or use the example sequence**

```
MVAFTEKQDALVSSSFEAFKANIPQYSVVFYTSILEKAPAAKDLFSFLANGVDPTNPKLTGHA EKLFALV
RDSAGQLKASGTVVADAALGVSVAQKAVTDPQFVVVKEALLKTIKAAVGDKWSDELSRAWEVAYDELA
IKKA
>rice_globin 1D8U rice Non-Symbiotic Plant Hemoglobin NP_001049476.1 [Oryza sativa (japonica cultivar-group)]
MALVEDNNAVAVSFSEEQALVLKSWAILKKDSANIALRFFLKIFEVAPSASQMFSFLRNSDVPLEKNPK
LKTHAMSVFVMTCEAAAQLRKAGKVTVRDTTLKRLGATHLKYGVGDAHFEVVKFALLDTIKEEVPADMWS
PAMKSAWSEAYDHLVAAIKQEMKPAE
```

[Choose File](#) [No file chosen](#) [Use the example](#) [Clear sequence](#) [More example inputs](#)

**Parameters**

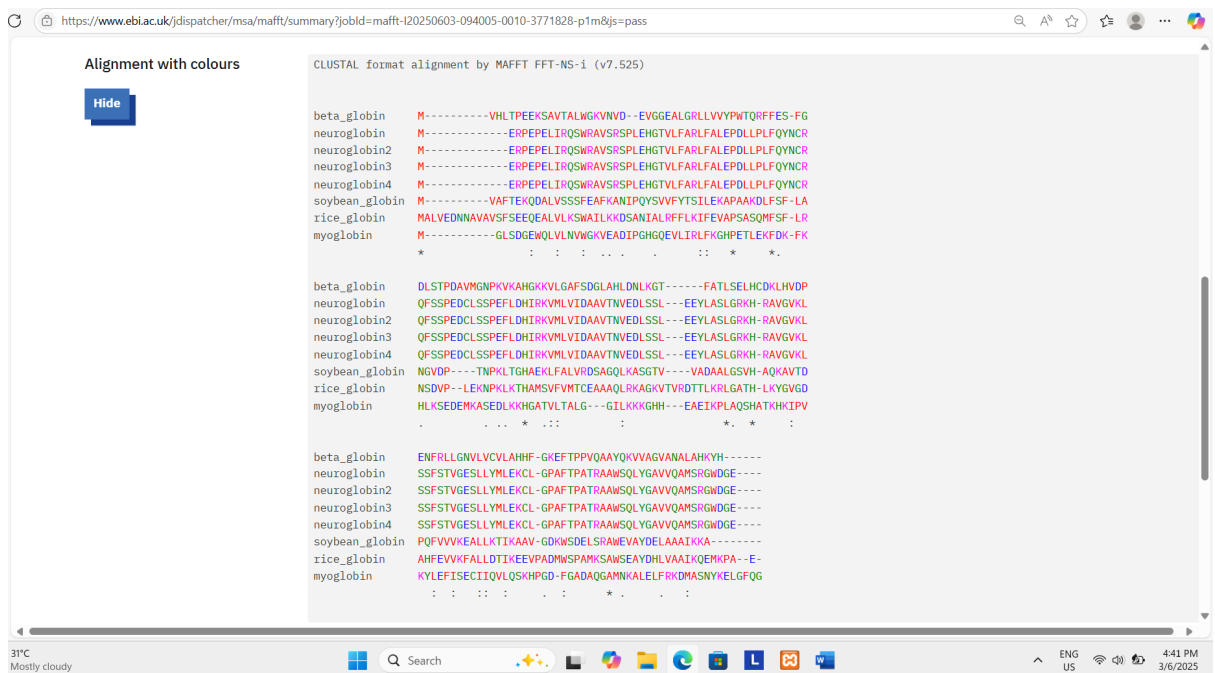
**OUTPUT FORMAT**   
ClustalW

[More options](#)

11°C Mostly cloudy 4:39 PM 3/6/2025



## The results:



Carefully inspect the region surrounding the main internal extension (of rice globin) and note that the alignment has now changed. Further note the gap in the first alignment, beginning of the middle block, included for soybean T--N. This region has been aligned quite differently in the second data set having added additional neuroglobins.

Given that this exercise showed that additional sequences changed the alignment, how can you decide which is “better”? One main answer is that we can use benchmarking to decide which alignments are better supported by structural information. Another possibility is to obtain a large set of homologs (e.g. 50 or 100 rather than 5) and gain stronger evidence of where the gap positions occur in nature. Another possibility is to use phylogeny-aware alignment strategies as discussed in the chapter.

## 5) Use the T-Coffee programs to evaluate the effect of structural information on your alignments. Follow these steps.

- Obtain a group of five distantly related lipocalins from web document 6.10 (► <http://www.bioinfbook.org/chapter6>). These include rat odorant-binding protein, and human retinol-binding protein.
- Align the sequences using T-Coffee (► <http://www.tcoffee.org/>) (WebLink 6.13), or use another program.
- Evaluate the alignment with the iRMSD program (► <http://www.tcoffee.org/>). Include the information on two known lipocalin structures. Note the score.

- iv. Align the same sequences again using Expresso (► <http://www.tcoffee.org/>) to incorporate structural information. Note the score. Did it improve? Do the alignments differ?

(5) SOLUTIONS/COMMENTS:

Here are eight lipocalins:

>human\_RBP4 gi|55743122|ref|NP\_006735.2| retinol-binding protein 4, plasma precursor [Homo sapiens]

MKWVWALLLLAALGSGRAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEGLFLQDNIVAEFVDETGG  
MSATAKGRVRLNNWDVCADMVGTFTDTPAKFKMKYWGVSFLQKGNDDHWIVDTDYDTYAVQYSCRL  
LNLDGTCADSYSFVSRDPNGLPPEAQKIVRQRQEELCLARQYRLIVHNGYCDGRSERNLL

>rat\_OBP gi|20302101|ref|NP\_620258.1| odorant binding protein I f [Rattus norvegicus]

MVKFLLIVLALGVSCAHHENLDISPSEVNGDWRTLYIVADNVEKVAEGGSLRAYFQHMECGDECEQELKII  
FNVKLDSECQTHTVVGQKHEDGRYTTDYSGRNYFHVLLKKTDDIIFHNVNVDSEGRRCDLVAGKREDLN  
KAQKQELRKLAEYNIPNENTQHLVPTDTCNQ

>NP\_006735 retinol-binding protein 4 [Homo sapiens]

MKWVWALLLLAALGSGRAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEGLFLQDNIVAEFVDETGG  
MSATAKGRVRLNNWDVCADMVGTFTDTPAKFKMKYWGVSFLQKGNDDHWIVDTDYDTYAVQYSCRL  
LNLDGTCADSYSFVSRDPNGLPPEAQKIVRQRQEELCLARQYRLIVHNGYCDGRSERNLL

>1QWD|A Bacterial Lipocalin Blc E. Coli

MSYYHHHHHHLESTSLYKSSSTPPRGVTVVNNFDAKRYLGTWYEIARFDHRFERGLEKVTATYSLRDDG  
GLNVINKGYNPDRGMWQQSEGKAYFTGAPTRAALKVSFFGPFYGGYNVIALDREYRHALVCGPDRDYLWI  
LSRTPTISDEVKQEMLA VATREGFDVSKFIWVQQPGS

>1Z24|A Chain A, The Molecular Structure Of Insecticyanin From The Tobacco Hornworm  
Manduca Sexta L. At 2.6 Å Resolution.

GDIFYPGYCPDVKPVNDFDLAFAAGAWHEIAKLPLENENQKCTIAEYKYDGKKASVYNSFVSNGVKEYM  
EGDLEIAPDAKYTKQGKYVMTFKFGQRVVNLVPWVLATDYKNYAINYNCDYHPDKKAHSIHAWILSKSKV  
LEGNTKEVVDNVLKTFSHLIDASKFISNDFSEAACQYSTTYSLTGPDRH

>2BLG Bovine Beta-Lactoglobulin

LIVTQTMKGLDIQKVAGTWYSLAMAASDISLLDAQSAPLRVYVEELKPTPEGDLEILLQKWENDECAQKK  
IIAEKTKIPAVFKIDALNENKVLVLDTDYKKYLLFCMENSAEPEQSLVCQCLVRTPEVDDEALEKFDKAL  
KALPMHIRLSFNPTQLEEQCHI

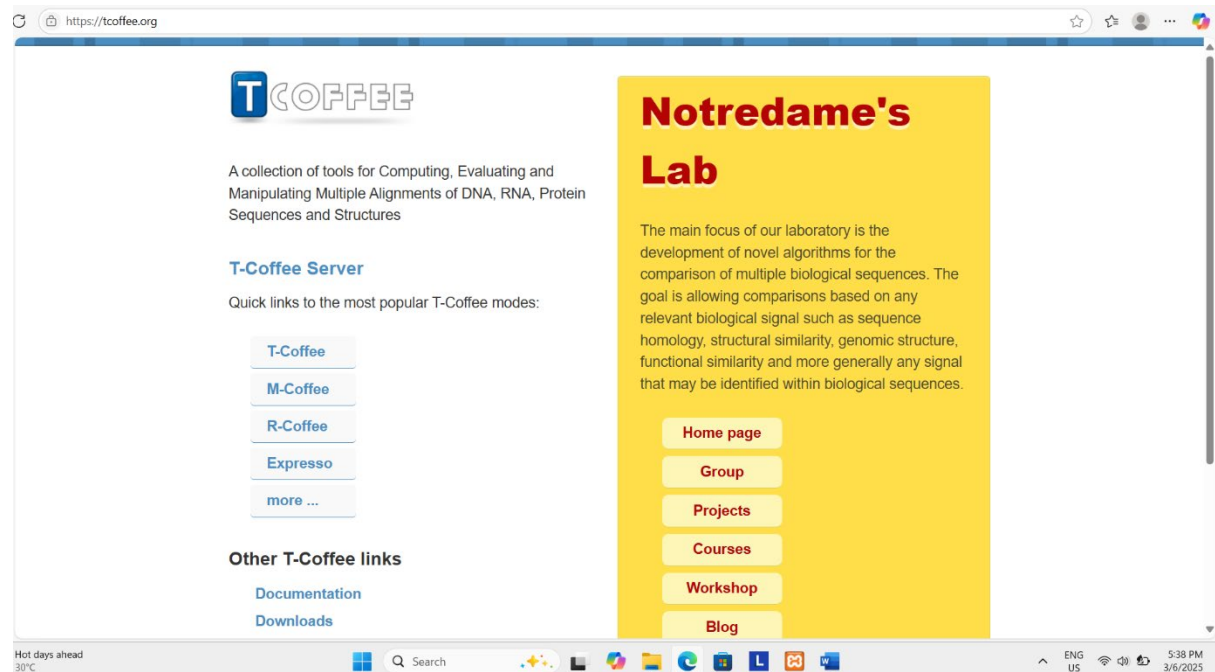
## >1PBO|A Bovine Odorant Binding Protein (Obp)

AQEEAEQNLSGLSGPWRTVYIGSTNPEKIQENGPFRTYFRELVFDDDEKGTVDYFYSVKRDGKWKNVHVK  
ATKQDDGTIVADYEGQNVFKIVSLRTHLVAHNINVDKHGQKTELTGLFVKLNVEDEDLEKFWKLTEDKG  
IDKKNVVNFLENEDHPHPE

## >1E5P|A Aphrodisin Female Hamster

QDFAELQGKWYTIVIAADNLEKIEEGGPLRFYFRHIDCYKNCSEXETFYVITNNQCSKTTVIGYLKNG  
TYETQFEGNNIFQPLYITSDKIFFTNKNXDRAGQETNXIVVAGKGNALTPEENEILVQFAHEKKIPVENI  
LNILATDTCPE

So , we are using website (<http://www.tcoffee.org/>) to analyse, and this is it homepage:



Put the sequences to analyse:

https://tcoffee.crg.eu/apps/tcoffee/dozregular

**T-COFFEE** Home History Tutorial References Contacts Projects Download

### T-Coffee

*Aligns DNA, RNA or Proteins using the default T-Coffee*

**Sequences input**  
Paste or upload your set of sequences in FASTA format

Sequences to align  
[Click here to use the sample file](#)

```
>1ESPIA Aphrodisin Female Hamster
AQEEAEQNLSELSGPWRTVYIGSTNPEKIQENGPFRTYRELVEDEKGTVDYFYSVKRDGKWKNNWVK
ATKQDDGTYVADYEGQNVFKIVSLSRTHLVAHNINVDKHGKTELTLGLFVKLNVEDLEKFWKLTEDKG
IDKKNVNFFLENDHPHE
>1ESPIA Aphrodisin Female Hamster
QDFAELQGWYTIIVIAADNLEKIEEGPLRFYFRHIDCYKNCSEXEIFYVITNNQCSKTTVIGYLGKNG
TYETQFEGNNIFQPLYITSDKIFFTNKNXDRAGQETNXIVVAGKGNALTPREENILVQFAHEKKIPVENI
LNILATDTCPE
```

- OR - [Click here to upload a file](#)

[Show more options](#)

Your email address

[Reset](#)

T-Coffee Server is hosted by the [Centre for Genomic Regulation](#) (CRG) of Barcelona - Powered by [nextflow](#) [Back to top](#)

30°C Mostly cloudy Search ENG US 5:40 PM 3/6/2025

The T-Coffee output:



6) This problem introduces multiple alignments in the MAF format at the Galaxy website.

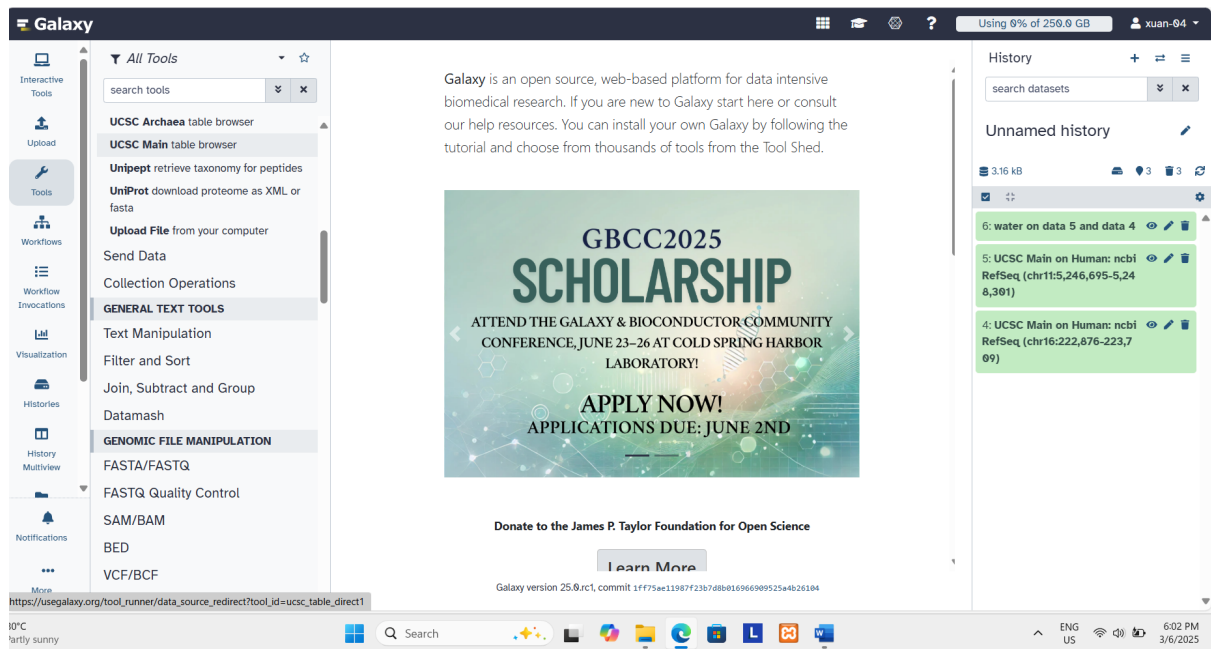
- i. Go to Galaxy (either the main public server or a local instance you set up).
- ii. Under the “Get Data” tool (left sidebar), select UCSC Main table browser. Choose mammal, human, Feb. 2009 (GRCh37/hg19), Genes and Gene Prediction Tracks (group), RefSeq Genes (track), and in the position box type in hbb then “lookup” to

obtain the coordinates chr11:5246696-5248301. Use the output format “BED” and send the output to Galaxy.

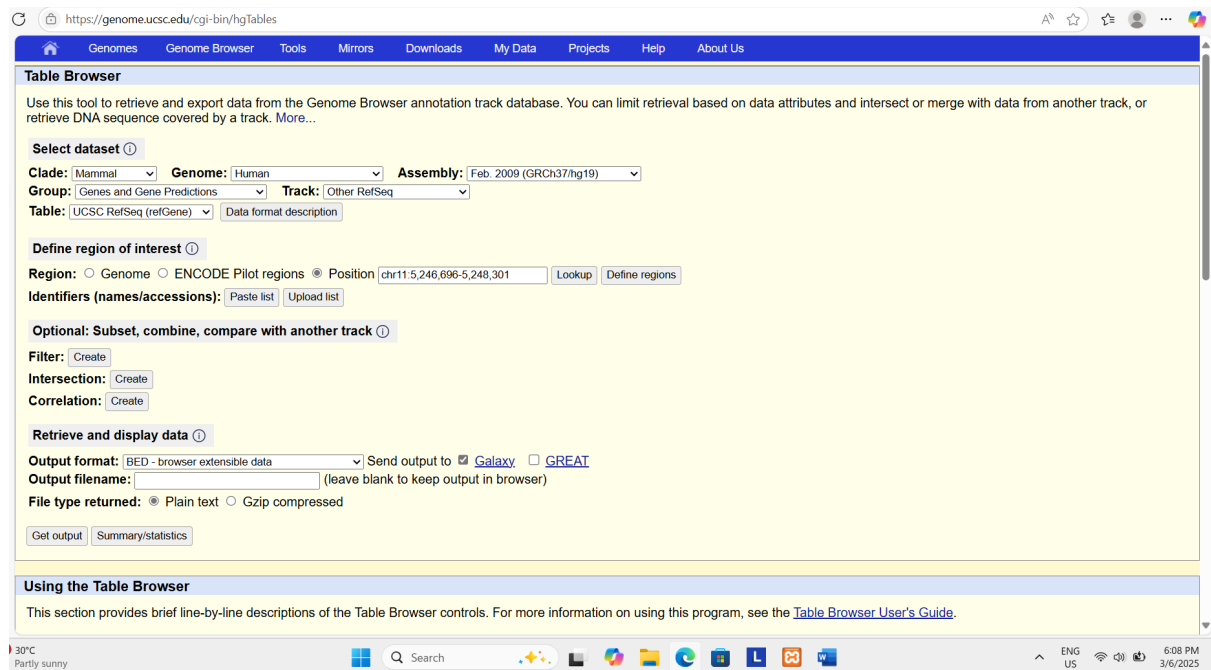
- iii. View the dataset.
- iv. Go to Tools > Fetch Alignments > Extract MAF blocks. For the interval, use the imported data set from the UCSC table browser; for MAF source, choose “Locally Cached Alignments”; and select “46-way multiZ (hg19).”
- v. Inspect the output (click the eye icon on the history panel). There are 37 blocks across this region. For each block, the line labeled “a” shows a float point score, while the lines labeled “s” correspond to sequences (these are 0-based starts).
- vi. On the tools panel choose “Graph/Display Data” then choose GMAJ. This is an interactive, Java-based multiple alignment viewer (Blanchette et al., 2004).
- vii. Convert the multiple alignment to a set of FASTA files. Go to Tools > Convert Formats > “MAF to FASTA” and choose one sequence per species as the type of FASTA output. Optionally you can download these sequences (e.g. to align them with different methods). For example, you can use Tools > Multiple Alignments > ClustalW to align these FASTA files.
- viii. Choose Tools > Evolution > “Neighbor Joining Tree Builder”. Use the FASTA file, and a distance model such as Kimura 2 parameter (see Chapter 7).
- ix. Choose Tools > Fetch Alignments > MAF Coverage Stats. Using the summarized coverage output option, the coverage is the number of nucleotides divided by total length of the given intervals.

#### (6) SOLUTIONS/COMMENTS:

Go to Galaxy website and at the left side click tool -> get data ->UCSC Table Browser:



The Table Browser settings are as follows; note the output is BED and the option “send output to Galaxy” is selected:



Select whole gene and send the output to Galaxy.

https://genome.ucsc.edu/cgi-bin/hgTables

Genomes Genome Browser Tools Mirrors Downloads My Data Projects Help About Us

### Output refGene as BED

☐ Include [custom track](#) header:

name=   
description=   
visibility=   
url=

Create one BED record per:

☒ Whole Gene  
☐ Upstream by  bases  
☐ Exons plus  bases at each end  
☐ Introns plus  bases at each end  
☐ 5' UTR Exons  
☐ Coding Exons  
☐ 3' UTR Exons  
☐ Downstream by  bases

Note: if a feature is close to the beginning or end of a chromosome and upstream/downstream bases are added, they may be truncated in order to avoid extending past the edge of the chromosome.

[Send query to Galaxy](#)  
[Cancel](#)

The output is green when ready; click it to expand, and click the eye icon to display information in the central display panel:

https://usegalaxy.org/datasets/f9cad7b01a472135ad5c9eb6c0c7d493

Galaxy

Using 0% of 250.0 GB xuan-04

Interactive Tools

search tools

Get Data  
Send Data  
Collection Operations

GENERAL TEXT TOOLS

Text Manipulation  
Filter and Sort  
Join, Subtract and Group  
Datamash

GENOMIC FILE MANIPULATION

FASTA/FASTQ  
FASTQ Quality Control  
SAM/BAM  
BED  
VCF/BCF  
Nanopore  
Convert Formats  
Lift-Over

COMMON GENOMICS TOOLS

7: UCSC Main on Human: refGene (chr11:5,246,696-5,248,301)

1 region 12 columns  
format bed database hg19

Preview Visualize Details Edit

| Chrom | Start   | End     | Name      | Score | Strand | ThickStart | ThickEnd | ItemRGB | BlockCount | BlockSizes  |
|-------|---------|---------|-----------|-------|--------|------------|----------|---------|------------|-------------|
| chr11 | 5246693 | 5248301 | NM_000518 | 0     | -      | 5246827    | 5248251  | 0       | 3          | 263,223,142 |

History

search datasets

Unnamed history

3.24 kB

7: UCSC Main on Human: refGene (chr11:5,246,696-5,248,301)

Hot days ahead 30°C

6:17 PM 3/6/2025

Follow the instructions to Tools > Fetch Alignments > Extract MAF blocks (on the left sidebar), the Execute.



Galaxy

Extract MAF blocks given a set of genomic intervals (Galaxy Version 1.0.1+galaxy1)

Tool Parameters

Choose intervals

7: UCSC Main on Human: refGene (chr11:5,246,696-5,246,391)

MAP Source

Locally Cached Alignments

Choose alignments

36-way multiZ (hg38)

Choose species

Select / Deselect all

☒ hg38

☐ panTro5

☐ panPan2

☐ gorGor5

☐ ponAbe2

☐ nomLeu3

☒ rhesMac8

☐ rhesMac5

☐ macMon1

☐ cerAty1

☐ papAnu3

☐ chSab2

☐ manLeu1

☐ narLar1

☐ colAng1

☐ rhiRox1

☐ rhiBie1

☐ calJac3

☐ salBoi1

☐ celCap1

☐ aotTan1

☐ tarSyr2

☐ micMur3

History

Unnamed history

7: UCSC Main on Human: refGene (chr11:5,246,696-5,246,391)

3.24 KB

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Here the output, there is no MAF blocks extracted

Galaxy

8: Extract MAF blocks on data 7

7 blocks format maf database hg19

No MAF blocks could be extracted for 1 regions.

Preview Visualize Details Edit

maf version=1

History

Unnamed history

8: Extract MAF blocks on data 7

Add Tags

7 blocks format maf database hg19

No MAF blocks could be extracted for 1 regions.

Species: maf version=1

7: UCSC Main on Human: refGene (chr11:5,246,696-5,246,391)

Add Tags

1 region 12 columns format bed database hg19

6:40 PM 3/6/2025